

DB-3.0

Definition Breaker

Playable by Me

A child-usable playbook and production standard for starting, understanding, choosing, repairing, pausing and returning - through physical toys, tabletop games and one shared digital world.

SEE

CHOOSE

CHECK

PLAY

PAUSE

RETURN



- Start**
one meaningful choice within the first minute
- Understand**
one stable word, shape and result for each state
- Repair**
undo, change route or say NOT THAT without penalty

Start playing in 90 seconds

No login, reading test or calibration is needed before the first real choice.



1

Show two choices
 Place or show two different rule tiles. Say or display what each one will do.

2

Choose a route
 The child may look, listen, touch, point, speak, sign, scan, press, gaze or use a partner.

3

Preview one change
 Move the choice into TRY. Show the likely result without changing the real world.

4

Confirm or repair
 Use YES, NO, NOT SURE or NOT THAT. Commit only after confirmation.

STOP

WAIT

HELP

UNDO

QUIET

BREAK

First-use rule

The first encounter must contain a safe success, a visible STOP route and an easy undo. It must not demand speed, speech, eye contact, fine motor control or reading.

My control bar is always here

These controls work before, during and after every action.

STOP**STOP**

Stop movement, sound, scanning and commitment now. Keep the current state safe.

WAIT**WAIT**

Give me more processing time. Do not repeat or add choices unless I ask.

HELP**HELP**

Explain, reduce clutter or offer another route. Do not choose for me.

UNDO**UNDO**

Return to the last accepted world state without shame or lost reward.

QUIET**QUIET**

Reduce sound, motion, glow, vibration, speech or social demand.

BREAK**BREAK**

Save where I am. Show a return card later instead of testing my memory.

Safety rule

No essential screen or physical activity may hide, disable or delay the control bar.

Choose and mix my routes

A route is how the world reaches me and how I answer. I can mix routes and change them.

LOOK

LISTEN

TOUCH

READ

SPEAK

SIGN

AAC

SWITCH

GAZE

HEAD

KEYS

PARTNER

A

Receive the world

Objects, relationships and states can be shown through words, symbols, sound, touch, objects, captions, narration or spatial layout.

B

Answer the world

A choice can arrive through movement, stillness, gaze, switch, speech, sign, AAC, keyboard, touch or confirmed partner relay.

Route-change rule

Changing route preserves the puzzle, candidate, reward, focus, scan position and last explanation.

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Child route

Every page repeats the same layout: three steps, a table version, a screen version, PLAY NOW, later vocabulary and the control bar.

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What is greatly improved in DB-3.0

DB-3.0 keeps the complete DB-2.0 access architecture and rebuilds the first-use experience around child usability.

1

A playable front book

Forty direct-use pages replace explanation-first onboarding. Each page has one purpose, three steps and one immediate activity.

2

Stable state words

TRY, CHECKING, READY, NOT YET, REAL and REMEMBERED replace vague success/failure messages.

3

Fewer choices at once

The default is two meaningful choices, with progressive disclosure when the child asks for more.

4

First success before setup

No account, calibration, tutorial quiz or reading test blocks the first real world change.

5

Child-controlled hints

Help climbs a visible ladder from reminder to solution; the child decides the level and can step back.

6

Session usability

One-action, 3-minute, 8-minute, 15-minute and open play routes support energy, attention and predictable endings.

7

Physical-digital sameness

Rule names, sockets, order rails, candidate states, commits and replay symbols transfer between toys and screens.

8

Usability evidence

New metrics measure start, choice, confirmation, repair, stop, resume, effort, joy and adult takeover.

Adult promise: help without taking over

The child remains the decision-maker even when another person reads, moves, scans, interprets or explains.

1**OFFER**

Show a small set of real choices in a route the child can receive.

2**WAIT**

Allow enough time. Do not fill silence with more instructions or repeat rapidly.

3**OBSERVE**

Notice speech, sign, gaze, movement, stillness, approach, withdrawal and individualized signals.

4**CONFIRM**

Repeat the interpreted choice and wait for YES, NO, NOT THAT or another agreed signal.

5**RELAY**

Move the piece or activate the control exactly as confirmed. Do not improve the child's choice.

6**REPAIR**

If the interpretation was wrong, stop, restore the prior state and try another route without blame.

Touch rule

Use touch only with consent. Prefer hand-under-hand exploration and "do with, not for" support where touch is part of an agreed route.

I am the world director

START

6 QUESTIONS

BUILD

USE

MAP



I CAN USE THIS IDEA

The world waits for my choice. A helper may relay a choice, but cannot replace it.

1

SEE

Find two different rule choices.

2

CHOOSE

Use any route that works today.

3

CHECK

Hear, see or feel the choice back before it happens.

ON THE TABLE

Place two large rule tiles on a clear mat. The child chooses one and the partner repeats the choice.

ON THE SCREEN

Focus moves to two named rule buttons. Selection is announced and remains a TRY until confirmed.

PLAY NOW

Choose between two cards, then use YES or NOT THAT.

My words

I decide what happens.

Later word

agency, proposal and confirmation

STOP

WAIT

HELP

UNDO

QUIET

BREAK

Pick a play time

- START
- 6 QUESTIONS
- BUILD
- USE
- MAP

TIME

I CAN USE THIS IDEA

I can choose a session that fits my energy. The end is visible before play starts.

- 1

SEE

Choose ONE ACTION, 3, 8, 15 or OPEN.
- 2

CHOOSE

Pick an end marker, not a pressure timer.
- 3

CHECK

Change the plan at any time without losing progress.

ON THE TABLE

Use duration cards and place FINISHED at the end of a short activity strip.

ON THE SCREEN

Choose a session card. The interface shows an effort forecast and saves after every accepted change.

PLAY NOW

Pick ONE ACTION or 3 MINUTES. Finish or change the plan when ready.

My words

I choose how long this play is.

Later word

session budget and checkpoint policy

- STOP
- WAIT
- HELP
- UNDO
- QUIET
- BREAK

Check the sensory forecast

START

6 QUESTIONS

BUILD

USE

MAP

FORECAST

I CAN USE THIS IDEA

I can know what sound, light, motion, touch, reading and people are likely before the room starts.

1

SEE

Review each forecast channel.

2

CHOOSE

Keep, reduce, skip or replace each channel.

3

CHECK

The room confirms the quiet and motion settings.

ON THE TABLE

Use cards for SOUND, LIGHT, MOTION, TOUCH, READING, PEOPLE and BREAKS.

ON THE SCREEN

A pre-room panel previews sound, motion and social demand without auto-playing them.

PLAY NOW

Choose one channel to keep and one to reduce or skip.

My words

Tell me what is coming.

Later word

sensory forecast and session access state

STOP

WAIT

HELP

UNDO

QUIET

BREAK

Choose one or more routes

START

6 QUESTIONS

BUILD

USE

MAP

ROUTE

I CAN USE THIS IDEA

I may combine seeing, listening, touch, reading, AAC, switches, gaze, keys or partner relay.

1

SEE

Find the route cards.

2

CHOOSE

Pick one or more routes.

3

CHECK

Try a sample choice and change anything that does not work.

ON THE TABLE

Place route cards beside the play mat. Add or remove them during play.

ON THE SCREEN

A route wizard tests output, input, confirmation and repair without changing difficulty.

PLAY NOW

Pick two routes that can work together.

My words

This is how the world reaches me.

Later word

multimodal access route

STOP

WAIT

HELP

UNDO

QUIET

BREAK

Pick a helper job

- START
- 6 QUESTIONS
- BUILD
- USE
- MAP

HELPER

I CAN USE THIS IDEA

I can decide what a helper may do and what stays mine.

- 1

SEE

Choose READER, MOVER, EXPLAINER or SCAN PARTNER.
- 2

CHOOSE

Mark what the helper may and may not do.
- 3

CHECK

The helper repeats the agreement.

ON THE TABLE

Give the helper one role card. The child keeps the COMMIT or CONFIRM card.

ON THE SCREEN

Co-play assigns permissions. The helper cannot commit, skip or change access settings without confirmation.

PLAY NOW

Choose one helper job and one thing the helper must not do.

- My words

Help me in the way I choose.
- Later word

supported decision-making role

- STOP
- WAIT
- HELP
- UNDO
- QUIET
- BREAK

Practice STOP and UNDO

START

6 QUESTIONS

BUILD

USE

MAP

STOP

I CAN USE THIS IDEA

I know how to stop and restore the last safe world before a hard puzzle begins.

1

SEE

Locate STOP and UNDO.

2

CHOOSE

Use STOP during a practice movement.

3

CHECK

Use UNDO and confirm that the prior state returns.

ON THE TABLE

Move a harmless piece, say STOP, then return it using the UNDO card.

ON THE SCREEN

Start a harmless preview animation, stop it, then restore the last committed snapshot.

PLAY NOW

Practice STOP once and UNDO once.

My words

I can stop and repair.

Later word

safe pause and rollback

STOP

WAIT

HELP

UNDO

QUIET

BREAK

Start with two choices

START

6 QUESTIONS

BUILD

USE

MAP

2

I CAN USE THIS IDEA

I get a small, meaningful choice set instead of a crowded wall of options.

1

SEE

Two choices are shown with clear effects.

2

CHOOSE

Select one, or ask for ANOTHER.

3

CHECK

More choices appear only when requested.

ON THE TABLE

Show two cards. Keep the rest face down or outside the active mat.

ON THE SCREEN

Display two primary choices plus ANOTHER and NOT SURE. Do not hide an important NO.

PLAY NOW

Choose A, B, ANOTHER or NOT SURE.

My words

Two is enough to start.

Later word

progressive disclosure and choice set

STOP

WAIT

HELP

UNDO

QUIET

BREAK

Hear or feel the choice back

START

6 QUESTIONS

BUILD

USE

MAP

CONFIRM

I CAN USE THIS IDEA

The system repeats what it thinks I chose before the real world changes.

1

SEE

The selected object and action are named.

2

CHOOSE

Answer YES, NO, NOT THAT or NOT SURE.

3

CHECK

Only YES moves to READY.

ON THE TABLE

The partner points to and names the chosen tile without moving it to the real mat.

ON THE SCREEN

The candidate is highlighted, spoken and summarized semantically; focus remains on confirmation controls.

PLAY NOW

Choose a tile and correct one deliberate wrong interpretation.

My words

Show me what you understood.

Later word

confirmation loop

STOP

WAIT

HELP

UNDO

QUIET

BREAK

Make one safe change

START

6 QUESTIONS

BUILD

USE

MAP

TRY

I CAN USE THIS IDEA

I can try a change in a candidate world before it becomes real.

1

SEE

Move the rule into TRY.

2

CHOOSE

Preview the result.

3

CHECK

Use COMMIT, CHANGE or CANCEL.

ON THE TABLE

Use a TRY mat beside the REAL WORLD mat. Nothing crosses the line without confirmation.

ON THE SCREEN

A preview layer shows affected objects, first failed check and likely result.

PLAY NOW

Make a candidate change, cancel it, then make a second one real.

My words

Try first. Make real later.

Later word

candidate proposal and verified commit

STOP

WAIT

HELP

UNDO

QUIET

BREAK

Pause and come back

START

6 QUESTIONS

BUILD

USE

MAP

RETURN

I CAN USE THIS IDEA

A break keeps my place and gives me a clear return summary instead of a memory test.

1

SEE

Use BREAK or close safely.

2

CHOOSE

Return now or later.

3

CHECK

Review YOU WERE HERE, LAST CHANGE and NEXT CHOICE.

ON THE TABLE

Clip a return card to the play mat with the current candidate and last committed move.

ON THE SCREEN

Restore focus, scan position, access route, candidate state and a concise scene summary.

PLAY NOW

Take a one-minute break and return using only the summary card.

My words

I can leave and still belong in the game.

Later word

checkpoint and resume contract

STOP

WAIT

HELP

UNDO

QUIET

BREAK

WHERE? Find the play area

START

6 QUESTIONS

BUILD

USE

MAP

WHERE

I CAN USE THIS IDEA

A rule works only in the places that support it.

1

SEE

Find the marked play area.

2

CHOOSE

Place the rule inside or outside it.

3

CHECK

The world says IN AREA or OUTSIDE AREA.

ON THE TABLE

Use a textured or bordered mat and one large rule tile.

ON THE SCREEN

A semantic region is named and highlighted; nonvisual output describes whether the rule is in support.

PLAY NOW

Find one place where the rule works and one where it does not.

My words

Where can this rule work?

Later word support

STOP

WAIT

HELP

UNDO

QUIET

BREAK

WHERE? Support mini-game

START

6 QUESTIONS

BUILD

USE

MAP



I CAN USE THIS IDEA

Support removes places that cannot matter before harder checking begins.

1

SEE

Three objects are near the rule.

2

CHOOSE

Keep only objects inside the support area.

3

CHECK

Explain by pointing, sorting, speaking, signing or scanning.

ON THE TABLE

Sort three objects into CAN REACH and CANNOT REACH baskets.

ON THE SCREEN

Use a support filter that leaves compatible checking for the next step.

PLAY NOW

Sort three objects. No speed score.

My words

First find what can reach.

Later word

support pruning

STOP

WAIT

HELP

UNDO

QUIET

BREAK

MATCH? Fit the right parts

START

6 QUESTIONS

BUILD

USE

MAP

MATCH

I CAN USE THIS IDEA

Two supported things still need compatible types, sheets, modes or permissions.

1

SEE

Compare sockets, words or tactile marks.

2

CHOOSE

Connect a matching pair.

3

CHECK

A mismatch explains which type did not match.

ON THE TABLE

Use distinct typed sockets and redundant word, shape and tactile cues.

ON THE SCREEN

Compatibility is exposed as semantic relation text, not only color or animation.

PLAY NOW

Connect one match and inspect one mismatch without penalty.

My words

Are these allowed to work together?

Later word compatibility

STOP

WAIT

HELP

UNDO

QUIET

BREAK

MATCH? Compatibility mini-game

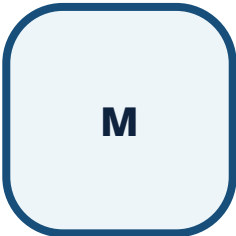
START

6 QUESTIONS

BUILD

USE

MAP



I CAN USE THIS IDEA

Being in the same place does not always mean two things can couple.

1

SEE

Place two pieces on the same spot.

2

CHOOSE

Change one sheet or type tag.

3

CHECK

Notice that position stays the same while compatibility changes.

ON THE TABLE

Use transparent or tactile overlays labeled SHEET A and SHEET B.

ON THE SCREEN

Toggle a compatibility tag while coordinates remain unchanged.

PLAY NOW

Make one same-place pair compatible and one incompatible.

My words

Same place can still be different.

Later word

compatibility-separated state

STOP

WAIT

HELP

UNDO

QUIET

BREAK

WHEN? Watch the gate

START

6 QUESTIONS

BUILD

USE

MAP

WHEN

I CAN USE THIS IDEA

A guard says when an important boundary or condition has really been reached.

1

SEE

Find the gate line or threshold.

2

CHOOSE

Move slowly toward it.

3

CHECK

The event wakes only when the declared rule is met.

ON THE TABLE

Slide a bead toward a raised threshold line with START and CROSSED sides.

ON THE SCREEN

A progress relation announces FAR, NEAR, IN BAND and CROSSED without flicker.

PLAY NOW

Move near the line, stop, then cross it.

My words

When is the change allowed?

Later word guard

STOP

WAIT

HELP

UNDO

QUIET

BREAK

WHEN? Guard mini-game

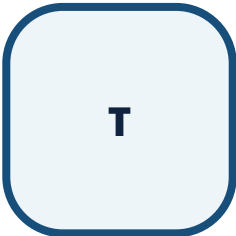
START

6 QUESTIONS

BUILD

USE

MAP



I CAN USE THIS IDEA

Near, touching and crossing are different states.

1

SEE

Use FAR, NEAR, TOUCH and CROSS cards.

2

CHOOSE

Match a movement to a state.

3

CHECK

A repeated wobble does not create endless changes.

ON THE TABLE

Use a wide tactile guard band and separate enter and exit marks.

ON THE SCREEN

Hysteresis and hold time prevent chatter; each state has text and nonvisual output.

PLAY NOW

Show TOUCH without CROSS, then CROSS once.

My words

Touching is not always crossing.

Later word

guard classification and hysteresis

STOP

WAIT

HELP

UNDO

QUIET

BREAK

CHECKED? Let the checker work

START

6 QUESTIONS

BUILD

USE

MAP

CHECKED

I CAN USE THIS IDEA

The world checks WHERE, MATCH and WHEN before it accepts a change.

1

SEE

The checker visits each question.

2

CHOOSE

Review the first check that is NOT YET.

3

CHECK

Repair that check or cancel.

ON THE TABLE

Move a checker token through WHERE, MATCH and WHEN stations.

ON THE SCREEN

A checklist exposes support, compatibility, guard and confidence without hidden failure.

PLAY NOW

Run one candidate through the three check stations.

My words

The world checks before changing.

Later word

verified event

STOP

WAIT

HELP

UNDO

QUIET

BREAK

CHECKED? Verified-event mini-game

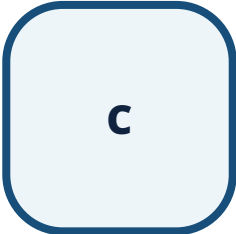
START

6 QUESTIONS

BUILD

USE

MAP



I CAN USE THIS IDEA
A checked change has a clear reason, not just a green light.

1

SEE
Compare a passing and a not-yet candidate.

2

CHOOSE
Ask WHY.

3

CHECK
The explanation names the first failed rule and a repair route.

ON THE TABLE
Use CHECKED and NOT YET cards with a removable reason strip.

ON THE SCREEN
Open the explanation panel from keyboard, switch, screen reader, touch or partner route.

PLAY NOW
Find and repair the first failed check.

My words
Tell me why it is ready or not yet.

Later word
verification reason code

STOP

WAIT

HELP

UNDO

QUIET

BREAK

CHANGE? Move TRY to REAL

START

6 QUESTIONS

BUILD

USE

MAP

CHANGE

I CAN USE THIS IDEA

A transition is the moment a checked candidate becomes the authoritative world.

1

SEE

READY appears after checking.

2

CHOOSE

Use COMMIT or CANCEL.

3

CHECK

The world names exactly what changed.

ON THE TABLE

Move the candidate tile across a clear TRY-to-REAL boundary.

ON THE SCREEN

Commit applies one atomic patch; cancellation leaves the real state unchanged.

PLAY NOW

Commit one ready change and cancel another.

My words

Make the checked change real.

Later word

transition and atomic patch

STOP

WAIT

HELP

UNDO

QUIET

BREAK

CHANGE? Transition mini-game

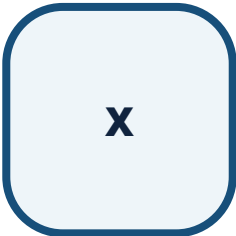
START

6 QUESTIONS

BUILD

USE

MAP



I CAN USE THIS IDEA
Only declared fields change. Everything else keeps its identity.

1

SEE
Mark the object name before change.

2

CHOOSE
Change one property.

3

CHECK
Confirm that object identity and untouched properties remain.

ON THE TABLE
Clip a new property card onto the same stable object card.

ON THE SCREEN
A before/after summary distinguishes node identity, changed fields and unchanged fields.

PLAY NOW
Change one property without replacing the object.

My words
The thing stays itself while one rule changes.

Later word
stable identity and transition patch

STOP

WAIT

HELP

UNDO

QUIET

BREAK

REMEMBER? Make a replay trail

START

6 QUESTIONS

BUILD

USE

MAP

REMEMBER

I CAN USE THIS IDEA

The world remembers accepted changes in order so I can replay, explain or return.

1

SEE

Start with a START bead.

2

CHOOSE

Add one bead per accepted change.

3

CHECK

Replay the sequence or stop at any bead.

ON THE TABLE

Thread large replay beads or arrange lineage cards left to right.

ON THE SCREEN

A replay list has sequence number, accessible summary, pre/post state and undo boundary.

PLAY NOW

Make a three-change trail and replay it.

My words

Remember what changed and in what order.

Later word

lineage and replay

STOP

WAIT

HELP

UNDO

QUIET

BREAK

REMEMBER? Lineage mini-game

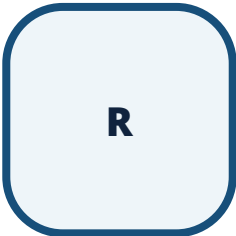
START

6 QUESTIONS

BUILD

USE

MAP



I CAN USE THIS IDEA
Two objects at the same place may still have different histories.

1

SEE
Give two identical pieces different replay trails.

2

CHOOSE
Place them on the same spot.

3

CHECK
Use the trail to tell them apart.

ON THE TABLE
Attach different lineage tags or bead strings to matching shapes.

ON THE SCREEN
Compare stable IDs and lineage summaries rather than coordinates alone.

PLAY NOW
Put two same-shape pieces together and identify each by its trail.

My words
History helps tell things apart.

Later word
lineage-based identity

STOP

WAIT

HELP

UNDO

QUIET

BREAK

A rule card has a name

START

6 QUESTIONS

BUILD

USE

MAP

RULE

I CAN USE THIS IDEA

A definition says what a rule is, what it needs, when it runs and what it may change.

1

SEE

Read or inspect the rule name and role.

2

CHOOSE

Open only the detail you need.

3

CHECK

The same rule name means the same thing on table and screen.

ON THE TABLE

Use one large tile with word, letter, border pattern and typed sockets.

ON THE SCREEN

A semantic rule object exposes name, role, state, dependencies and available actions.

PLAY NOW

Make a rule card with one purpose and one clear effect.

My words

A rule is a named recipe.

Later word definition

STOP

WAIT

HELP

UNDO

QUIET

BREAK

Some rules need other rules

START

6 QUESTIONS

BUILD

USE

MAP

NEEDS

I CAN USE THIS IDEA

A dependency says which definition or object must exist first.

1

SEE

Find the NEEDS socket.

2

CHOOSE

Connect the required rule.

3

CHECK

Missing needs are named before play starts.

ON THE TABLE

Use cables or labeled connectors between rule cards.

ON THE SCREEN

A dependency graph rejects missing references and cycles before execution.

PLAY NOW

Connect one required rule and remove it to see the clear NOT YET reason.

My words

This rule needs that rule.

Later word

dependency

STOP

WAIT

HELP

UNDO

QUIET

BREAK

First and then can change the result

START

6 QUESTIONS

BUILD

USE

MAP

ORDER

I CAN USE THIS IDEA

Operation order is part of the rule world. The same cards can make different results in a different order.

1

SEE

Place two rule cards on FIRST and THEN.

2

CHOOSE

Preview the result.

3

CHECK

Swap order and compare.

ON THE TABLE

Use an operation-order rail with stable numbered positions.

ON THE SCREEN

A phase-ordered pipeline records exact rule order and produces a trace.

PLAY NOW

Compare MOVE then TURN with TURN then MOVE.

My words

First and then matter.

Later word

non-commutative composition order

STOP

WAIT

HELP

UNDO

QUIET

BREAK

A hinge connects without adding size

START

6 QUESTIONS

BUILD

USE

MAP

HINGE

I CAN USE THIS IDEA

A connector can have an important structural role even when it adds no numeric amount.

1

SEE

Find the connector between two parts.

2

CHOOSE

Insert or remove it.

3

CHECK

Notice how the graph or order changes.

ON THE TABLE

Use a connector tile labeled AND, EN, JOIN or SWITCH.

ON THE SCREEN

A hinge node maps between charts, orders or rule branches while preserving declared invariants.

PLAY NOW

Insert a connector that changes order but not numeric value.

My words

A small connector can change the structure.

Later word

hinge

STOP

WAIT

HELP

UNDO

QUIET

BREAK

Same place can be a different sheet

START

6 QUESTIONS

BUILD

USE

MAP

SHEET

I CAN USE THIS IDEA

Coordinates are not the whole identity. A sheet label can keep overlapping worlds separate.

1

SEE

Stack SHEET A and SHEET B.

2

CHOOSE

Place one piece on each sheet at the same spot.

3

CHECK

Test whether the pieces are compatible.

ON THE TABLE

Use distinct textured overlays or raised sheet frames.

ON THE SCREEN

The scene query reports coordinate, sheet, phase and compatibility separately.

PLAY NOW

Place two objects at one coordinate but on different sheets.

My words

Same spot, different world layer.

Later word

sheet state and topology

STOP

WAIT

HELP

UNDO

QUIET

BREAK

A flip can change orientation

START

6 QUESTIONS

BUILD

USE

MAP

FLIP

I CAN USE THIS IDEA

Some wraps return an object mirrored or orientation-reversed.

1

SEE

Mark an arrow on the object.

2

CHOOSE

Move it through the flip track.

3

CHECK

Compare the arrow before and after.

ON THE TABLE

Use a track with raised entry/exit arrows and an orientation token.

ON THE SCREEN

A topology transition applies an explicit coordinate and orientation map.

PLAY NOW

Send an arrow tile through the flip track once and twice.

My words

A route can bring me back flipped.

Later word

orientation-reversing gluing

STOP

WAIT

HELP

UNDO

QUIET

BREAK

The same spot can have a new loop count

START

6 QUESTIONS

BUILD

USE

MAP

LOOP

I CAN USE THIS IDEA

Phase can repeat while winding remembers how many full loops happened.

1

SEE

Move around a loop.

2

CHOOSE

Add one loop bead each time the start is crossed.

3

CHECK

Compare the same spot with different bead counts.

ON THE TABLE

Use a circular track and large winding counters.

ON THE SCREEN

The state stores phase and integer winding separately.

PLAY NOW

Stop at the start after one loop and after two loops.

My words

Same place on the loop, different journey count.

Later word

phase and winding lineage

STOP

WAIT

HELP

UNDO

QUIET

BREAK

A small seed can rebuild a bounded world

- START
- 6 QUESTIONS
- BUILD
- USE
- MAP

SEED

I CAN USE THIS IDEA

A short seed may select a declared world recipe, but the recipe and limits must also be known.

1

SEE

Choose a seed card.

2

CHOOSE

Build using the matching recipe.

3

CHECK

Rebuild the same bounded result later.

ON THE TABLE

Use seed cards plus a visible rule deck and piece budget.

ON THE SCREEN

A versioned seed selects deterministic branches inside explicit grammar and resource limits.

PLAY NOW

Build the same small world twice from one seed card.

My words

A small code can choose a known recipe.

Later word

seeded deterministic reconstruction

- STOP
- WAIT
- HELP
- UNDO
- QUIET
- BREAK

A growing recipe needs a stop budget

START

6 QUESTIONS

BUILD

USE

MAP

GROW

I CAN USE THIS IDEA

A procedural grammar can branch and grow, but it must stop at a declared depth, symbol or piece limit.

1

SEE

Choose a growth depth.

2

CHOOSE

Apply one production at a time.

3

CHECK

Stop at the budget and count the result.

ON THE TABLE

Use a bounded branch deck with DEPTH 1, 2, 3 and STOP cards.

ON THE SCREEN

The runtime rejects depth, symbol, stack or time-budget overflow.

PLAY NOW

Grow a branch pattern to depth 2 and stop exactly at the budget.

My words

Grow, but stop safely.

Later word

bounded grammar

STOP

WAIT

HELP

UNDO

QUIET

BREAK

A short code must keep the needed meaning

START

6 QUESTIONS

BUILD

USE

MAP

CODE

I CAN USE THIS IDEA

A smaller record is only useful when it keeps the meaning and error needed for the task.

1

SEE

Compare a full card and a short code.

2

CHOOSE

Rebuild or query from the short code.

3

CHECK

Name what was kept and what was lost.

ON THE TABLE

Use paired FULL and SHORT cards with a checklist of retained facts.

ON THE SCREEN

Compression metrics state reconstruction, error and semantic limits instead of only bit ratios.

PLAY NOW

Find one short code that rebuilds correctly and one that loses an important fact.

My words

Shorter is not the same unless it keeps what matters.

Later word

honest compression boundary

STOP

WAIT

HELP

UNDO

QUIET

BREAK

Ask for a clue, not a takeover

START

6 QUESTIONS

BUILD

USE

MAP

HINT

I CAN USE THIS IDEA

I choose how much help I receive and may ask for less, more or a different route.

1

SEE

Open the hint ladder.

2

CHOOSE

Pick a level.

3

CHECK

The clue names what it changed and leaves the final decision mine.

ON THE TABLE

Use five hint cards from REMIND GOAL to SHOW SOLUTION.

ON THE SCREEN

Help is a child-controlled ladder and never auto-commits a move.

PLAY NOW

Use one small hint, then choose whether to climb or return.

My words

Help me without taking my choice.

Later word

scaffolded hint policy

STOP

WAIT

HELP

UNDO

QUIET

BREAK

Choose a challenge dial

START

6 QUESTIONS

BUILD

USE

MAP

LEVEL

I CAN USE THIS IDEA

Challenge, access and help are separate. I can make the puzzle deeper without giving up my access route.

1

SEE

Choose EXPLORE, BUILD, EXPLAIN or DESIGN.

2

CHOOSE

Keep the access routes that work.

3

CHECK

Rewards remain equal for equal concept outcomes.

ON THE TABLE

Use a four-position challenge dial beside the access-route cards.

ON THE SCREEN

Difficulty changes concept depth or number of relations, not caption, switch, AAC or hint availability.

PLAY NOW

Play the same concept once in EXPLORE and once in BUILD.

My words

Make the idea deeper, not harder to access.

Later word

separate challenge and access axes

STOP

WAIT

HELP

UNDO

QUIET

BREAK

Use any way to show what I know

START

6 QUESTIONS

BUILD

USE

MAP

SHOW

I CAN USE THIS IDEA

I may demonstrate understanding by acting, sorting, building, pointing, explaining, replaying or teaching a partner.

1

SEE

Choose a response route.

2

CHOOSE

Show the concept.

3

CHECK

The observer records the concept, not speech or motor style.

ON THE TABLE

Offer build, sort, point, act, AAC and partner-relay response cards.

ON THE SCREEN

Assessment accepts equivalent semantic outcomes from multiple input and output routes.

PLAY NOW

Show WHERE? in two different ways.

My words

There are many valid ways to answer.

Later word

multiple means of action and expression

STOP

WAIT

HELP

UNDO

QUIET

BREAK

Play together with equal jobs

START

6 QUESTIONS

BUILD

USE

MAP

TEAM

I CAN USE THIS IDEA

Cooperative roles have equal value and do not rank speech, speed, vision or hand use above other routes.

1

SEE

Choose roles such as OBSERVER, MATCHER, GUARD KEEPER or REPLAY GUIDE.

2

CHOOSE

Share or swap roles.

3

CHECK

Every role can use any access route.

ON THE TABLE

Use cooperative role cards and one shared COMMIT card controlled by the agreed child authority.

ON THE SCREEN

The multiplayer system separates role permission from input device and provides accessible shared summaries.

PLAY NOW

Complete one mini-game with two roles, then swap.

My words

Different jobs, equal importance.

Later word

cooperative role parity

STOP

WAIT

HELP

UNDO

QUIET

BREAK

Repair is part of winning

- START
- 6 QUESTIONS
- BUILD
- USE
- MAP

REPAIR

I CAN USE THIS IDEA

A misunderstanding or wrong move is useful information, not a punishment or lost status.

- 1

SEE

Name what the system thought happened.
- 2

CHOOSE

Use NOT THAT, UNDO or CHANGE ROUTE.
- 3

CHECK

Restore the last safe state and continue.

ON THE TABLE

Use the repair ladder and move the piece back without removing earned progress.

ON THE SCREEN

Error messages distinguish child intent, interpretation, rule failure and system fault.

PLAY NOW

Make one deliberate mismatch and repair it calmly.

My words

Fixing the route is part of the game.

Later word

repairable interaction and rollback

- STOP
- WAIT
- HELP
- UNDO
- QUIET
- BREAK

Rewards do not depend on speed or speech

- START
- 6 QUESTIONS
- BUILD
- USE
- MAP

FAIR

I CAN USE THIS IDEA

I earn progress for understanding, building, explaining or cooperating - not for being fastest, speaking aloud or working without support.

- 1

SEE

Choose a celebration route.
- 2

CHOOSE

Finish when the concept is shown.
- 3

CHECK

No access tool lowers score, ending or role.

ON THE TABLE
Offer quiet, visual, tactile, verbal or no-celebration choices.

ON THE SCREEN
Scoring excludes input speed, speech, gaze behavior and assistance status unless explicitly teaching a chosen timing skill.

PLAY NOW
Choose a preferred celebration and complete without a timer.

My words
My route is not a cheat.

Later word
outcome and dignity parity

- STOP
- WAIT
- HELP
- UNDO
- QUIET
- BREAK

Choose the next adventure

START

6 QUESTIONS

BUILD

USE

MAP

NEXT

I CAN USE THIS IDEA

I decide whether to repeat, review, try a new concept, save, take a break or finish.

1

SEE

Review what was learned in one sentence or object cue.

2

CHOOSE

Pick AGAIN, REVIEW, NEW, SAVE, BREAK or FINISH.

3

CHECK

The game accepts FINISH without pressure.

ON THE TABLE

Use a six-choice ending board, reduced to two choices when needed.

ON THE SCREEN

A predictable end screen preserves state and provides multiple ways to leave or continue.

PLAY NOW

Choose AGAIN or FINISH after one microgame.

My words

I choose what comes next.

Later word

session transition and learner agency

STOP

WAIT

HELP

UNDO

QUIET

BREAK

My play recipe

START

6 QUESTIONS

BUILD

USE

MAP

MY

I CAN USE THIS IDEA

My preferred routes, controls, session size and help style can be saved privately and changed at any time.

1

SEE

Review MY YES, MY NO, MY STOP and WHAT HELPS.

2

CHOOSE

Share only what I want.

3

CHECK

The profile stays separate from score, difficulty and public identity.

ON THE TABLE

Complete a portable access passport with words, symbols, objects or partner-supported observation.

ON THE SCREEN

Store a private AccessProfile and temporary AccessSessionState with child-controlled sharing and deletion.

PLAY NOW

Choose one route, one control signal, one help preference and one return preference.

My words

This is how play works best for me today.

Later word

access profile and session state

STOP

WAIT

HELP

UNDO

QUIET

BREAK

Communication is authority

A game action begins with the child's expressed intent.
Communication access therefore belongs inside the authoritative input path, not in a side note.

1

Accept many forms

Speech, AAC, sign, tactile sign, object cue, eye gaze, switch, facial expression and individualized movement can all carry intent.

2

Confirm meaning

The system or partner repeats the interpreted choice before commit.

3

Preserve repair

NOT THAT and CHANGED MIND remain available after every interpretation.

System rule

The proposal record stores the selected action and confirmation status, not a diagnosis or judgement about communication ability.

Communication before symbols

Some children communicate before or without conventional symbols. Orientation, approach, withdrawal, stilling, vocalization, breathing change or repeated movement may carry preference.

1

Observe over time

A single movement is not automatically assigned a meaning.

2

Offer real contrasts

Present two genuinely different, safe outcomes.

3

Check consistency

Confirm with people who know the child and remain open to change.

System rule

Pre-symbolic communication requires careful supported decision-making. It must not become mind reading.

Child-specific YES and NO

YES and NO may look different for each child and may change with posture, fatigue or context.

1

Define together

Record the child's current reliable signals.

2

Test neutrally

Use known-answer questions only to verify the route, not to interrogate the child.

3

Allow uncertainty

NOT SURE and NO RESPONSE stay distinct.

System rule

A communication passport may store a description of signals with the child's permission.

No response is not consent

Silence, stillness, delay, looking away, sleep, shutdown or tracking loss must not be treated as YES.

1

Pause

Stop the pending action.

2

Check access

Ask whether the child can perceive and operate the route.

3

Offer choices

WAIT, BREAK, DIFFERENT ROUTE or FINISHED.

System rule

No-response state is explicit in the input model and cannot auto-commit.

AAC core vocabulary

Core words let a child control the process across many rooms.

1

Control

STOP, GO, WAIT, AGAIN,
HELP, FINISHED.

2

Decision

YES, NO, NOT SURE, NOT
THAT, CHANGED MIND.

3

Access

QUIET, DIM, STILL,
BIGGER, READ, TOUCH,
SWITCH ROUTE.

System rule

Core words remain in stable locations so motor and visual learning can develop.

AAC fringe vocabulary

Puzzle-specific words sit beside the core without replacing it.

1

Rule words

SUPPORT, MATCH,
GUARD, PREVIEW,
COMMIT, REPLAY.

2

Object words

TILE, CABLE, GATE,
TRACK, SHEET, SEED.

3

Explanation words

WHERE, WHY, FIRST
FAILED CHECK, WHAT
CHANGED.

System rule

Fringe words use the same stable semantic IDs as onscreen rule objects.

Aided language modeling

Adults may point to or activate AAC words while speaking, without demanding imitation.

1

Model naturally

Use the board during real choices.

2

Do not test

Avoid “show me” drills during a decision.

3

Honor output

When the child uses a word, respond to its meaning.

System rule

Modeling supports access; it does not authorize an adult to select on the child's behalf.

Partner-assisted scanning

A partner presents choices in a predictable order while the child signals when to stop or select.

1

Agree signals

Define YES, NO, STOP and AGAIN.

2

Name neutrally

Use the same wording and pace each pass.

3

Confirm

Repeat the selected item before acting.

System rule

The scan order, group structure and confirmation result can be logged for replay and quality review.

Communication-partner fidelity

The partner is a relay, not a hidden controller.

1

Choice fidelity

Did the committed action match the child's confirmed intent?

2

Pace fidelity

Did the partner wait long enough?

3

Repair fidelity

Was NOT THAT honored immediately?

System rule

Partner fidelity is a release and training metric, not a judgement of the child.

Interpretation repair

Misunderstandings are expected and repairable.

1

Pause

Do not continue from an uncertain interpretation.

2

State the guess

"I thought you chose
GUARD."

3

Offer correction

YES, NO, NOT THAT,
SHOW AGAIN or
DIFFERENT ROUTE.

System rule

The world changes only after repaired confirmation.

Touch consent

Touch support begins with permission and stops at the child's stop signal.

1

Tell before touch

Name who, where and why.

2

Offer alternatives

Object cue, verbal direction, model hand or hand-under-hand.

3

Stop immediately

Withdrawal or STOP ends contact.

System rule

Surprise touch and forced hand movement are prohibited.

Hand-under-hand exploration

The adult's hand acts as a platform or model beneath the child's hand, allowing the child to join, leave or lead.

1

Invite

Offer the object and position.

2

Follow

Let the child control pressure and duration.

3

Describe

Name the object and relation without forcing a response.

System rule

This is “do with,” not “do for.” It requires individualized specialist guidance when used in deafblind education.

Object cues

A real object or consistent miniature can announce an activity before it starts.

1

Stable meaning

The same cue always means the same activity.

2

Tactile access

Shape and texture are distinguishable.

3

Choice use

Offer object cues for PLAY, BREAK and FINISHED.

System rule

Object cues may connect to the same stable activity IDs used digitally.

Tactile schedules

A left-to-right or top-to-bottom sequence of objects shows FIRST, THEN, NEXT and FINISHED.

1

Explore first

Let the child inspect the schedule.

2

Move the marker

The child or partner marks completed steps.

3

Allow change

A changed plan is explained and physically updated.

System rule

The schedule is a prediction aid, not a behavior-compliance contract.

Sign language and interpreters

Signed communication receives equal time, sightlines and authority.

1

Layout

Keep the signer, interpreter and relevant object visible.

2

Timing

Pause visual action while critical signed explanation occurs.

3

Terminology

Agree signs for substrate terms and preserve them consistently.

System rule

Captions do not replace sign language for every child; sign and text can be offered together.

Multilingual communication

The child may understand rules best in a home language, sign language, symbol system or mixed mode.

1

Separate stable ID from label

Translations change the displayed phrase, not the rule identity.

2

Allow text expansion

Layouts reflow for longer words.

3

Check meaning

Use community review for key concepts and safety words.

System rule

The system must never infer lower ability from language choice.

Communication fatigue

Expressing a choice can require more energy than understanding it.

1

Reduce repetition

Do not make the child restate a confirmed choice.

2

Offer shortcuts

Use stored phrases and recent-choice history with permission.

3

Pause scanning

Allow rest without losing focus.

System rule

Effort is measured separately from correctness.

Group communication

Cooperative play needs turn access that is visible and negotiable.

1

Role cards

Observer, Matcher, Guard
Keeper, Committer and
Replay Guide.

2

Pass or pause

A child may pass without
losing status.

3

Shared confirmation

Group choices show
whose confirmation is
still needed.

System rule

No role is reserved for speaking, reading or physically moving pieces.

Privacy and dignity

Communication preferences can be private and shared only as needed.

1

No public diagnosis label

Show a route, not a medical category.

2

No spectacle

Do not announce that access tools are “special.”

3

Child control

Let the child choose what appears in the Access Passport.

System rule

Access data stays separate from score, behavior ratings and advertising profiles.

Emergency communication

STOP, PAIN, HELP, TOO MUCH, NEED SPACE and CONTACT MY PERSON must remain reachable even when ordinary play is paused.

1

Always available

Do not bury safety words in a submenu.

2

Multiple channels

Provide visible, spoken, tactile and switch-accessible routes.

3

Human response

Define who receives the signal and what they do next.

System rule

Emergency communication does not depend on completing the current scan or puzzle step.

Design for functions, not labels

A diagnosis never predicts one fixed route. Ask what the child needs to perceive, understand, act, confirm, repair, regulate and return.

1

Requirement 1

Offer route combinations.

2

Requirement 2

Let settings change during the session.

3

Requirement 3

Test with the child, not on assumptions.

Parity test

Record supports and barriers, not a deficit score.

Blind access

Build a navigable semantic graph with concise and detailed scene summaries, landmarks, relations, actions and change announcements.

1

Requirement 1

No coordinate-only instructions.

2

Requirement 2

No visual-only preview.

3

Requirement 3

Replay names objects and relations.

Parity test

A blind player can inspect a rule, its dependencies, compatibility and candidate result without using a pixel map.

Low-vision access

Support large text, reflow, magnification, high contrast, strong focus, uncluttered views and stable target positions.

1

Requirement 1

Do not force zoom-out.

2

Requirement 2

Keep text selectable.

3

Requirement 3

Preserve contrast in all states.

Parity test

Magnification does not remove access to commit, undo or error explanation.

Cerebral visual impairment (CVI)

Provide low-complexity views, controlled movement, strong object separation, familiar anchors, extra visual latency and nonvisual equivalents.

1

Requirement 1

One object at a time.

2

Requirement 2

Reduce background detail.

3

Requirement 3

Allow preferred visual field and viewing distance.

Parity test

CVI support is individualized; the system offers adjustable visual complexity rather than a single "CVI mode."

Color-vision access

Color can reinforce meaning but never carry it alone.

1

Requirement 1

Add word labels.

2

Requirement 2

Add shape and border pattern.

3

Requirement 3

Add position or tactile code.

Parity test

Every state difference remains identifiable in monochrome.

Deaf access

Provide captions, sign-compatible layouts, visible timing, text transcripts and visual equivalents for all audio cues.

1

Requirement 1

Do not cover the signer with UI.

2

Requirement 2

Pause competing visual action during key signed information.

3

Requirement 3

Let captions name speaker and sound source.

Parity test

Rhythm and timing cues have visible or tactile forms.

Hard-of-hearing access

Allow independent speech, music and effect levels; clear captions; replay; and optional hearing-device-friendly audio settings.

1

Requirement 1

No whisper-only clue.

2

Requirement 2

No alert hidden in music.

3

Requirement 3

Repeat without penalty.

Parity test

Audio is supportive, never the only route to rule state.

Deafblind access

Use stable tactile orientation, object cues, tactile symbols, hand-under-hand exploration, tactile schedules and partner-confirmed communication.

1**Requirement 1**

Tell before touch.

2**Requirement 2**

Keep a tactile home position.

3**Requirement 3**

Represent change through object movement or replay beads.

Parity test

Every critical tactile interaction follows consent and individualized communication practice.

Motor access

Avoid precision, sustained force, rapid repetition and mandatory simultaneous actions.

1

Requirement 1

Large targets.

2

Requirement 2

Adjustable snap and dwell.

3

Requirement 3

Toggle, step and partner-assisted alternatives.

Parity test

Motor effort is not part of puzzle difficulty unless explicitly chosen by the child.

One-hand access

Every two-hand action has a sequential or toggle route.

1

Requirement 1

Anchor then transform.

2

Requirement 2

Lock one axis at a time.

3

Requirement 3

Use mode cycling instead of chords.

Parity test

The conceptual relation remains the same even when bimanual input is serialized.

Switch access

Provide one-switch auto-scan, two-switch next/select, step scan, grouping, back, pause and adjustable timing.

**1****Requirement 1**

Stable order.

**2****Requirement 2**

Visible and spoken focus.

**3****Requirement 3**

No timeout after selection.

Parity test

Switch scanning reaches every essential action, including settings, help, undo and exit.

Gaze and head access

Use large stable targets, rest zones, adjustable dwell, smoothing, confirmation and safe tracking-loss behavior.

1

Requirement 1

No gaze-only irreversible commit.

2

Requirement 2

Avoid edge-crowded targets.

3

Requirement 3

Keep a nonmoving rest area.

Parity test

Tracking loss pauses and preserves focus instead of interpreting movement as a choice.

Tremor, ataxia and dyspraxia

Support movement filtering, target expansion, snap, undo, delayed activation and non-drag routes.

1

Requirement 1

Do not punish repeated entries.

2

Requirement 2

Offer discrete direction controls.

3

Requirement 3

Separate hover from select.

Parity test

Accidental activation is repairable without score loss.

Speech, AAC and nonspeaking access

No essential action, explanation or social role requires speech.

1

Requirement 1

Named actions map to AAC.

2

Requirement 2

Text entry has symbol/choice alternatives.

3

Requirement 3

Voice recognition is optional.

Parity test

Communication route does not alter reward or story status.

Cognitive and intellectual disability access

Use concrete language, stable routines, visible cause and effect, one decision at a time and meaningful repetition.

1

Requirement 1

Show before asking.

2

Requirement 2

Use real examples.

3

Requirement 3

Avoid trick wording.

Parity test

The formal concept is preserved while explanation depth and pacing are adjustable.

Reading and dyslexia access

Offer narration, plain language, short lines, generous spacing, symbol support and a no-reading route.

1

Requirement 1

Do not use all caps for paragraphs.

2

Requirement 2

Avoid dense justified text.

3

Requirement 3

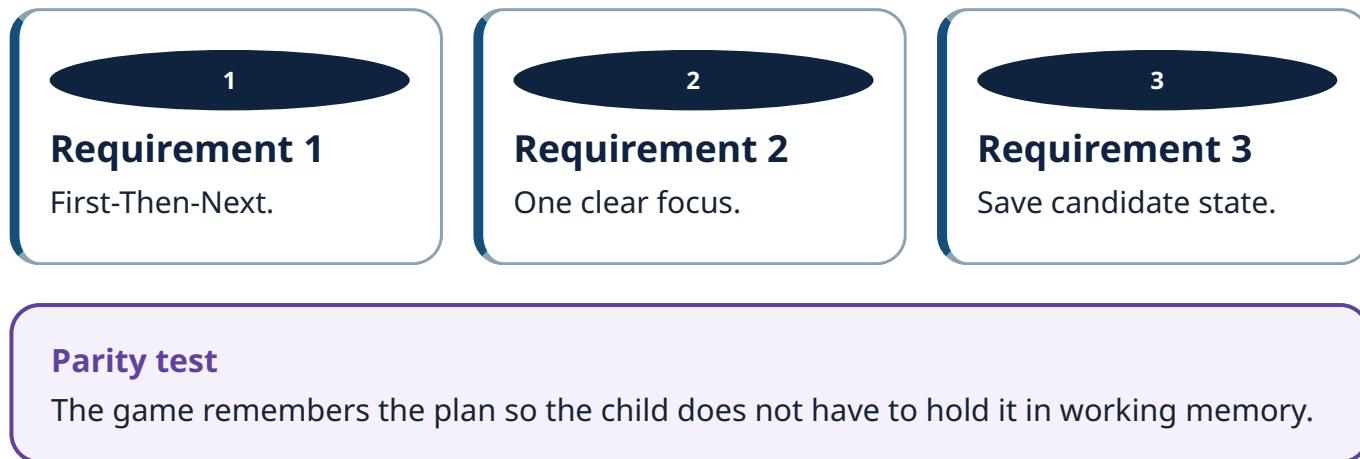
Keep words selectable and reflowable.

Parity test

Reading skill is not a gate to topology or rule reasoning.

Executive-function access

Externalize sequence, status, next action, unfinished choices and return point.



Autistic and sensory access

Provide predictability, direct language, sensory controls, interests as optional bridges, and no forced eye contact or social performance.

1

Requirement 1
Preview transitions.

2

Requirement 2
Offer quiet/still routes.

3

Requirement 3
Allow solo or structured co-play.

Parity test
Regulation is access, not a reward for compliance.

Attention variability

Use short chunks, resumable goals, persistent focus, low-distraction mode and optional movement breaks.

1

Requirement 1

No rapid reminder spam.

2

Requirement 2

Let the child mark “come back later.”

3

Requirement 3

Keep important state visible.

Parity test

Attention changes do not erase understanding or decision authority.

Photosensitivity and migraine

Avoid flashing, high-contrast flicker, rapid luminance change and unavoidable particle effects; provide a low-stimulation preset before first play.

1

Requirement 1

No flashing essential clue.

2

Requirement 2

Motion and glow can be disabled.

3

Requirement 3

Pause immediately.

Parity test

Safety settings are available before any potentially triggering content.

Vestibular and motion sensitivity

Offer static camera, reduced parallax, no camera shake, slower transitions and nonanimated alternatives.

1

Requirement 1

No forced head motion.

2

Requirement 2

No zoom pulse.

3

Requirement 3

Provide teleport or step transitions where relevant.

Parity test

Reduced motion retains event timing and state information.

Fatigue, pain and chronic illness

Support micro-sessions, save anywhere, low-force input, effort forecasts, quiet pauses and return summaries.

1

Requirement 1
No streak loss.

2

Requirement 2
No repeated calibration.

3

Requirement 3
No access setup tax each session.

Parity test
Energy varies; the child can change route without losing the world.

Trauma-informed and emotional safety

Use clear boundaries, predictable consequences, opt-out routes, nonpunitive repair and no forced disclosure.

1

Requirement 1
Avoid surprise restraint or touch.

2

Requirement 2
Do not simulate helplessness as onboarding.

3

Requirement 3
Keep exit and pause visible.

Parity test
The game never uses access needs as horror, punishment or moral failure.

Combined access

Test routes in real combinations: screen reader plus switch, low vision plus gaze, AAC plus partner scan, deafblind plus fatigue.

1

Requirement 1

Features must coexist.

2

Requirement 2

Focus and captions must not conflict.

3

Requirement 3

One route must not disable another.

Parity test

Single-feature compliance is insufficient when children use combined access.

Fluctuating access and device loss

Access may change within minutes. A device disconnect, tracking loss or overload must pause safely and preserve the same object, candidate and confirmation state.

1

Requirement 1

Offer another route.

2

Requirement 2

Explain what was preserved.

3

Requirement 3

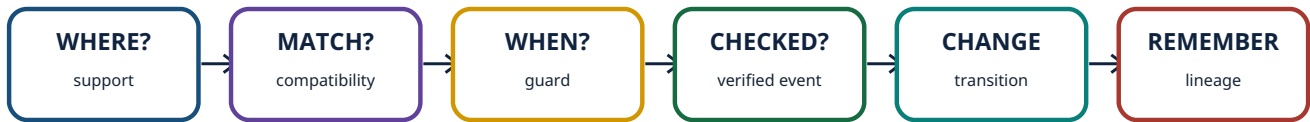
Resume without recalibration when possible.

Parity test

Temporary access session state is saved separately from authoritative puzzle state.

A world made of rules

Definition Breaker turns invisible rules into objects that can be noticed, moved, connected, checked and remembered.



Child words

Definition Breaker turns invisible rules into objects that can be noticed, moved, connected, checked and remembered.

Grown-up bridge

A door can carry its own support, compatibility and guard cards. A child can inspect the reason it opens instead of memorizing a hidden script.

Formal substrate

Definitions, instances, relations, supports, compatibility gates, events, transitions and lineage are first-class substrate objects.

Play prompt

Ask the child to show the idea with a tile, word, sound, gesture, switch or partner-assisted choice.

Why questions came first

The substrate began by asking what needs to be known before a change is allowed.

Child words

The substrate began by asking what needs to be known before a change is allowed.

Grown-up bridge

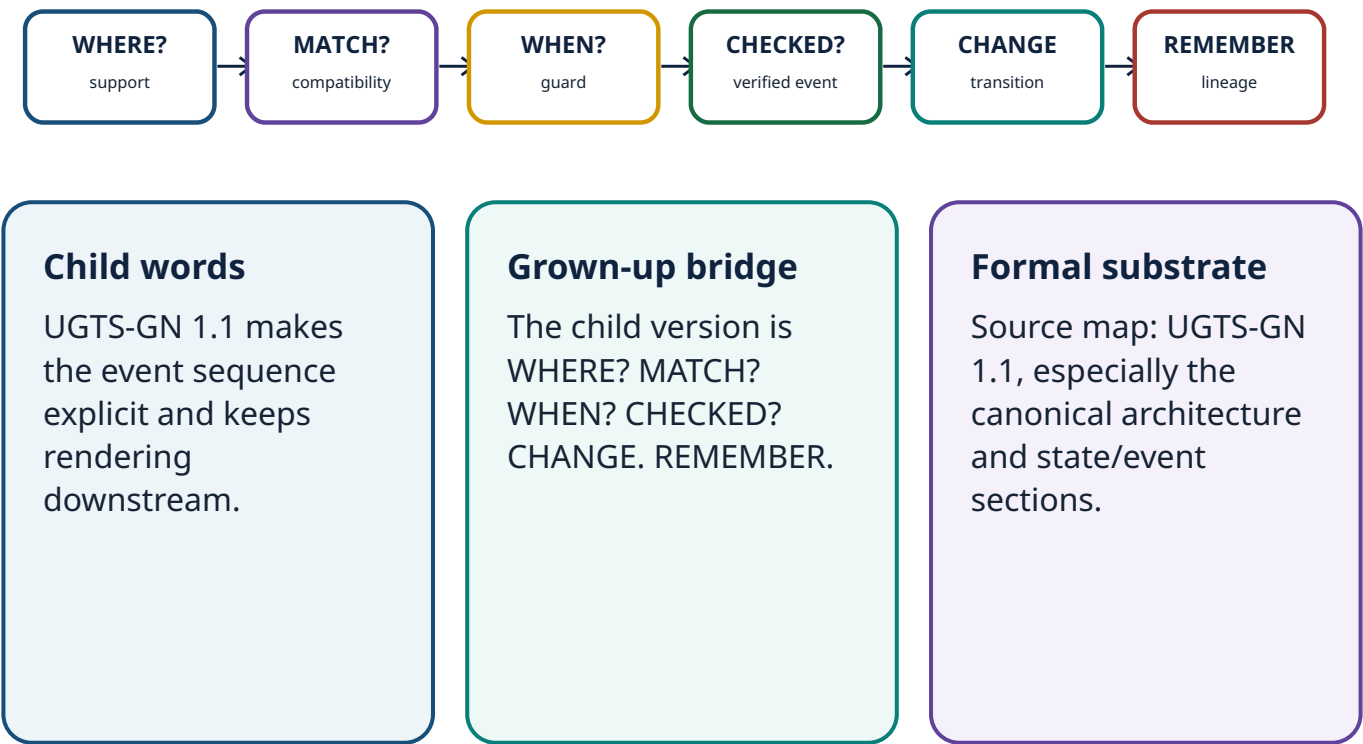
First find what is nearby and relevant. Then ask whether it may interact. Then check a boundary. Only then may the world change.

Formal substrate

The canonical query-first authority chain is support -> compatibility -> guard -> verified event -> transition -> lineage.

GN 1.1: the six-question backbone

UGTS-GN 1.1 makes the event sequence explicit and keeps rendering downstream.



KC 2.0: patterns, fields and motion

KC 2.0 adds shape families, implicit fields, route identity, moving frames, bounded dynamics and hybrid events.



Child words

KC 2.0 adds shape families, implicit fields, route identity, moving frames, bounded dynamics and hybrid events.

Grown-up bridge

Children can meet these as path cards, field mats, motion arrows and “same place, different journey” stories.

Formal substrate

Patterns and dynamics extend the operator set but do not replace support, compatibility, guards or lineage.

Two Hands 3.0: interaction, proposals and replay

Two Hands 3.0 adds stable scene nodes, hand interaction, candidate proposals, deterministic commit, checkpoints and replay.

Child words

Two Hands 3.0 adds stable scene nodes, hand interaction, candidate proposals, deterministic commit, checkpoints and replay.

Grown-up bridge

A child may grab a rule tile, preview the effect, confirm, and replay the accepted change.

Formal substrate

Input systems may propose in parallel; only verified deterministic commit changes authoritative state.

KC 3.6: definitions live inside the world

KC 3.6 turns definitions, dependencies, phases and operation order into inspectable records.

Child words

KC 3.6 turns definitions, dependencies, phases and operation order into inspectable records.

Grown-up bridge

A rule card can say what it is, what it needs, when it runs and which invariant it must keep.

Formal substrate

Each literal definition has stable ID, type, dependencies, phase, invariants, provenance and content hash.

SCLP 3.6.2: bounded worlds and winding

SCLP adds guard intervals, phase plus winding, reflective topology, constraint release, bounded grammar and finite key layouts.

Child words

SCLP adds guard intervals, phase plus winding, reflective topology, constraint release, bounded grammar and finite key layouts.

Grown-up bridge

Children can count loops, flip a track, remove one shackle, grow a finite branching plant and compare seed cards.

Formal substrate

The profile is finite and query-first; compression claims require reconstruction and error equivalence.

KC 3.9: a playable vector runtime

KC 3.9 adds vector assets, named multi-device actions, fixed-step simulation, collisions, animation, sound, persistence and offline HTML5.

Child words

KC 3.9 adds vector assets, named multi-device actions, fixed-step simulation, collisions, animation, sound, persistence and offline HTML5.

Grown-up bridge

The accessible game can use the same project data for keyboard, gamepad, touch, switch adapters and deterministic replay.

Formal substrate

One versioned GameProject can validate, simulate and build to self-contained browser output.

SARA 3.6.3: public, private and permitted

For older learners, SARA shows that typed information can be public, private or permitted only under an authorization guard.

Child words

For older learners, SARA shows that typed information can be public, private or permitted only under an authorization guard.

Grown-up bridge

A public label does not give permission to discover or use a secret. Some paths must be rejected.

Formal substrate

This is an optional ethics and security lesson. Secret search, signing, broadcast and funds transfer stay outside children's play.

The six big questions

Every rule change can be explained through six short questions.



Child words

Every rule change can be explained through six short questions.

Grown-up bridge

WHERE can it work?
MATCH may these things work together?
WHEN did the boundary happen?
CHECKED is the evidence good enough?
CHANGE what becomes different?
REMEMBER what must be kept?

Formal substrate

These are the child-language bridge to support, compatibility, guard, verified event, transition and lineage.

Play prompt

Ask the child to show the idea with a tile, word, sound, gesture, switch or partner-assisted choice.

Definitions

A definition says what kind of rule something is and how it behaves.

Child words

A definition says what kind of rule something is and how it behaves.

Grown-up bridge

A SUPPORT tile might define a circle around a sensor. Every tile with that definition can use the same rule.

Formal substrate

A literal definition record declares ID, kind, domain, codomain, parameters, dependencies, phase, invariants and provenance.

Stable identity

A thing can move or change appearance without becoming a different thing.

Child words

A thing can move or change appearance without becoming a different thing.

Grown-up bridge

The blue bridge tile keeps its identity when the player rotates it or changes its material.

Formal substrate

Node identity, asset identity, coordinates and lineage are distinct.

Dependencies

A dependency says what a rule needs before it can work.

Child words

A dependency says what a rule needs before it can work.

Grown-up bridge

The OPEN rule may need a KEY rule and a MATCH rule. Pulling the OPEN card also shows those needed cards.

Formal substrate

Definition dependencies form an acyclic graph and are resolved before evaluation.

Operation order

Doing the same actions in a different order can make a different world.

Child words

Doing the same actions in a different order can make a different world.

Grown-up bridge

MOVE then ROTATE is not always the same as ROTATE then MOVE. The order rail makes this visible.

Formal substrate

Composition is generally noncommutative; the evaluation schedule is explicit and traceable.

Hinges

A hinge connects, pivots or selects a map. It can matter structurally even when it adds no number.

Child words

A hinge connects, pivots or selects a map. It can matter structurally even when it adds no number.

Grown-up bridge

A connector tile can join UNIT, EN and TENS. A parity hinge can choose route A or B.

Formal substrate

A hinge carries pivot or connector locus, state, selected map, guard and invariants.

Support

Support asks whether a rule is relevant here.

Child words

Support asks whether a rule is relevant here.

Grown-up bridge

A sound gate may work only inside the blue circle. Outside the circle, it is not considered.

Formal substrate

Local spatial support prunes irrelevant candidates before expensive relation work.

Compatibility

Compatibility asks whether supported things are allowed to couple.

Child words

Compatibility asks whether supported things are allowed to couple.

Grown-up bridge

Two tokens may share a square but belong to different sheets, teams or phases, so they do not interact.

Formal substrate

Compatibility uses sheet, phase, orientation, mode, policy, ownership and application tags.

Guards

A guard watches for a declared boundary or condition.

Child words

A guard watches for a declared boundary or condition.

Grown-up bridge

A door guard may wait until the counter reaches three. Near the boundary, it can report NOT SURE rather than guess.

Formal substrate

Guard classification uses a relation, margin, uncertainty and crossing/touch/tangency policy.

Candidate world

A candidate world is a safe “what if?” state.

Child words

A candidate world is a safe “what if?” state.

Grown-up bridge

The child connects a new rule and sees the predicted route without changing the real board.

Formal substrate

Candidate proposals remain nonauthoritative until support, compatibility, guard and policy checks pass.

Verified proposal

A proposal becomes verified only when every required check passes with enough confidence and small enough error.

Child words

A proposal becomes verified only when every required check passes with enough confidence and small enough error.

Grown-up bridge

The game can say: "It is in range, it matches, the guard crossed, and the evidence is clear."

Formal substrate

verified = support_ok
and compatibility_ok
and accepted_guard
and confidence $\geq$
floor and
numeric_error $\leq$
margin.

Commit and change

Commit applies one checked patch to the real world.

Child words

Commit applies one checked patch to the real world.

Grown-up bridge

The candidate bridge becomes the real bridge only after the child confirms COMMIT.

Formal substrate

Verified proposals are ordered, conflicts are resolved by versioned policy, and one authoritative patch is applied.

Lineage

Lineage remembers where an object or rule came from and how it changed.

Child words

Lineage remembers where an object or rule came from and how it changed.

Grown-up bridge

A split token becomes child A and child B. A merged object records both parents.

Formal substrate

Durable identity uses generative address, transition history, invariants and novelty records rather than coordinates alone.

Replay

Replay rebuilds a world from a start state, ordered events and checkpoints.

Child words

Replay rebuilds a world from a start state, ordered events and checkpoints.

Grown-up bridge

The child can hear or feel: "You moved MATCH, previewed, committed, then changed the route."

Formal substrate

Each event carries sequence and pre/post hashes; replay stops at the first divergence.

Coordinates are not identity

Two things can be at the same place and still be different.

Child words

Two things can be at the same place and still be different.

Grown-up bridge

A bridge on sheet A and a bridge on sheet B overlap in the picture but need not connect.

Formal substrate

Position equality does not imply coupling or identity; sheet, route, orientation and lineage remain separate.

Sheets and phases

A sheet or phase is an extra state label that separates worlds sharing the same coordinates.

Child words

A sheet or phase is an extra state label that separates worlds sharing the same coordinates.

Grown-up bridge

A red phase token can pass through a blue phase wall until a MATCH rule changes.

Formal substrate

Sheet and phase are typed compatibility fields, not hidden visual effects.

Topology and orientation

Topology describes how places are connected and how orientation changes across a boundary.

Child words

Topology describes how places are connected and how orientation changes across a boundary.

Grown-up bridge

A track can wrap so LEFT becomes RIGHT after one trip. The transition is shown by an arrow and flip token.

Formal substrate

Möbius or Klein behavior is an explicit gluing map with coordinate, sheet, orientation and lineage transport.

Winding remembers loops

Phase can repeat while winding remembers how many full loops happened.

Child words

Phase can repeat while winding remembers how many full loops happened.

Grown-up bridge

Two tokens at the same clock position can have winding 1 and winding 4, so their histories differ.

Formal substrate

Linear time maps to periodic phase plus integer winding lineage.

Patterns

A pattern is a versioned shape or path with parameters and a declared role.

Child words

A pattern is a versioned shape or path with parameters and a declared role.

Grown-up bridge

Rose, spiral, clothoid and Gielis cards can make routes, supports or puzzle boundaries.

Formal substrate

Pattern instances declare mechanism ID, parameters, frame, support, compatibility tags and validation contract.

Fields

A field gives a value at each place. Zero can be a boundary, but not every field is an exact distance.

Child words

A field gives a value at each place. Zero can be a boundary, but not every field is an exact distance.

Grown-up bridge

A heat mat can say negative inside, zero at the edge and positive outside.

Formal substrate

Implicit-field sign, gradient and guard semantics are separate from exact signed-distance capability.

Motion and frames

Motion says where an object is and how its local directions move.

Child words

Motion says where an object is and how its local directions move.

Grown-up bridge

A car on a curved track has forward, side and up directions that change along the path.

Formal substrate

Kinematic queries return `state_at(t)`, `frame`, `curvature` and `bounded status`; a frame loop is not always required.

Constraints

A constraint keeps a relation true over an interval.

Child words

A constraint keeps a relation true over an interval.

Grown-up bridge

A token clipped to a rail can move along it but not leave it.

Formal substrate

Resting contact, joints and cooperative grabs are maintained constraints, not infinite zero-crossing events.

The missing shackle

Removing one independent constraint can add one legal direction of motion.

Child words

Removing one independent constraint can add one legal direction of motion.

Grown-up bridge

Take away one bar from a sliding frame and test the new movement. Do not call the result chaos automatically.

Formal substrate

Constraint-row deletion changes rank and nullity; dynamics require a separate mass, force and damping model.

Bounded grammar and branches

A grammar grows a structure using declared productions and strict budgets.

Child words

A grammar grows a structure using declared productions and strict budgets.

Grown-up bridge

A seed card grows a branching path for four levels, then stops. Every push has a matching pop.

Formal substrate

Depth, symbol count, stack and branch density are finite and overflow is rejected.

Seeds, keys and honest compression

A seed can rebuild regular structure, while true outside changes still need event records.

Child words

A seed can rebuild regular structure, while true outside changes still need event records.

Grown-up bridge

A short world code may reconstruct a maze, but it cannot magically contain infinite detail.

Formal substrate

Finite keys have declared ranges and quantization; width ratios are not equivalent compression without semantic and error equivalence.

Capstone: explain a full rule change

Use the whole path to explain one puzzle solution.

Child words

Use the whole path to explain one puzzle solution.

Grown-up bridge

WHERE: the bridge is in the bay. MATCH: the key and lock agree. WHEN: the counter crossed three. CHECKED: uncertainty is below the margin. CHANGE: the bridge opens. REMEMBER: replay records the accepted rule.

Formal substrate

A complete trace names definitions, dependencies, evaluation order, support, compatibility, guard status, proposal, commit patch and lineage.

Play prompt

Ask the child to show the idea with a tile, word, sound, gesture, switch or partner-assisted choice.

One semantic world

All input and output routes operate on the same semantic objects and authoritative state.

1

One object ID

Visual tile, spoken label, tactile symbol and AAC word refer to the same rule ID.

2

One proposal path

Keyboard, switch, gaze and partner input produce the same named action proposal.

3

One outcome

Verification, reward, story consequence and replay are route-independent.

Implementation contract

Do not fork an “accessible game” with different rules unless the child deliberately chooses a different difficulty.

AccessProfile stays separate

Store access preferences separately from score, achievement, puzzle difficulty and authoritative world state.

1

Private by default

The child chooses what is saved and shared.

2

No diagnosis inference

Settings do not become a guessed medical label.

3

Portable when chosen

A child may export preferences without exporting gameplay history.

Implementation contract

AccessProfile may include presentation, input, communication, language and regulation preferences.

AccessSessionState is temporary

Keep current scan focus, gaze rest state, caption position, communication route and sensory level as resumable session state.

1

Preserve interruption

Pause, device loss and route change keep state.

2

Expire safely

Temporary details can clear when the child chooses.

3

Never score it

Session access state is not performance data.

Implementation contract

Separate session state prevents a switch or gaze route from contaminating puzzle identity.

Named actions

Gameplay consumes stable intent names rather than device-specific buttons.

1

Examples

MOVE, INSPECT,
PREVIEW, COMMIT,
UNDO, STOP, HELP,
BREAK.

2

Multiple bindings

Keyboard, gamepad,
touch, switch, gaze and
partner adapters can
bind to the same action.

3

Full menus

Settings and shell
controls use named
actions too.

Implementation contract

KC 3.9 already uses device-independent action maps; DB-3.0 extends this to access and communication actions.

Input adapters

Adapters translate device signals into named action values and edge events without changing rule semantics.

1

Validate range

Clamp, debounce and report confidence.

2

Expose status

Tracking unavailable and device disconnected are states, not silent failure.

3

Replay intent

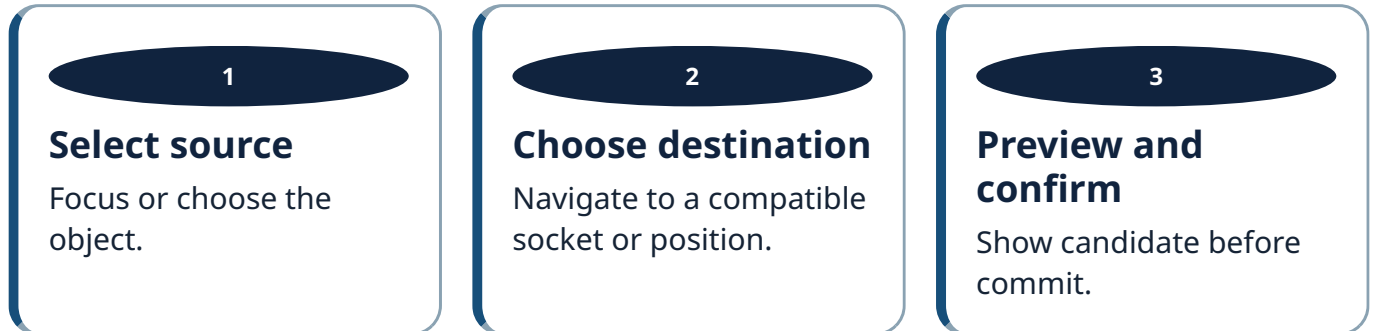
Record resolved actions, not raw biometric streams by default.

Implementation contract

Assistive input remains downstream of child intent and upstream of proposal verification.

No-drag route

Every drag has select-destination, directional step, snap, scan or partner-assisted equivalent.



Implementation contract

The no-drag route reaches the same placement precision through discrete semantics.

No-hold route

Every sustained hold has toggle, timed auto-complete, latch or separate start/stop equivalent.

1

Requirement 1

Hold to inspect -> toggle inspect.

2

Requirement 2

Hold to charge -> choose amount then confirm.

3

Requirement 3

Hold and move -> anchor then move sequentially.

Implementation contract

Holding strength or duration is not required unless explicitly chosen as the challenge.

Remapping and conflict detection

All named actions can be remapped, and the system warns when bindings collide.

1

Accessible before play

Remapping works on first launch.

2

Reset and undo

A safe default is always recoverable.

3

Per context

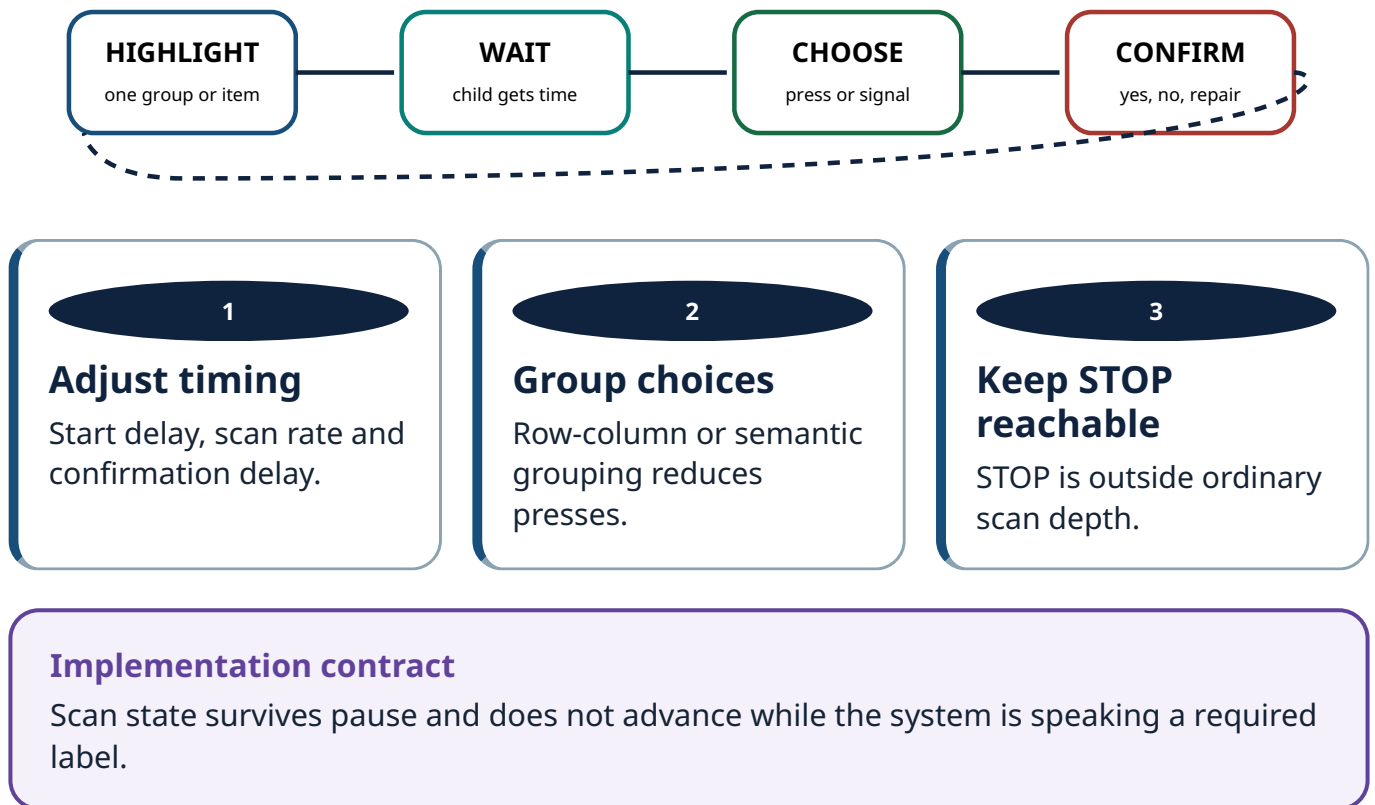
Menu and gameplay conflicts are explained.

Implementation contract

Do not reserve important keys or buttons without an alternative.

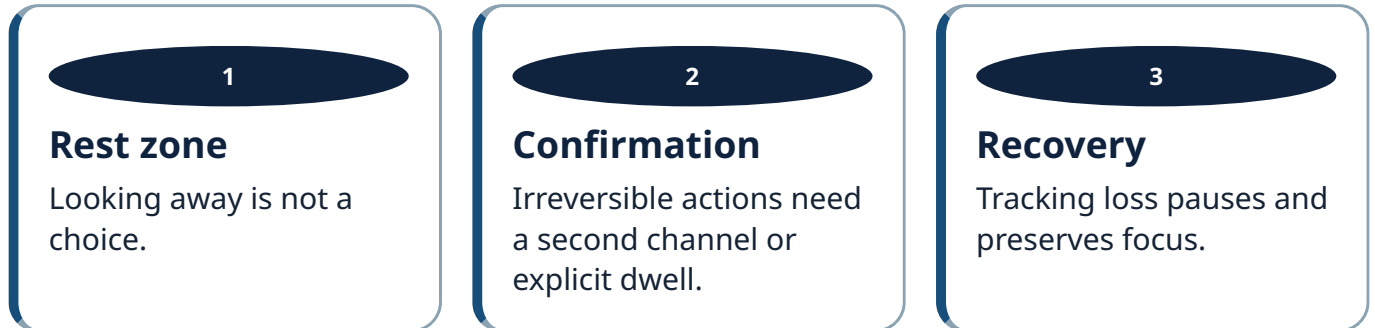
Switch scanning state machine

Scanning has explicit highlight, wait, choose, confirm, back and pause states.



Gaze state machine

Gaze navigation separates pointing, dwell, rest, confirmation and tracking-loss states.



Implementation contract

Raw gaze paths are not stored by default.

Predictable focus

Focus moves in a stable logical order and returns to the initiating object after dialogs, help or settings.

1

Visible and spoken

Focus has high-contrast and screen-reader state.

2

No focus trap

Every modal has cancel and return.

3

No surprise reset

Filtering or route change preserves the nearest semantic focus.

Implementation contract

Focus order follows semantic structure, not arbitrary screen coordinates.

Screen-reader object contract

Every rule object exposes name, role, state, value, description, dependencies, relations, available actions and change history.

1 Name "Compatibility seal B."	2 State "Candidate, disconnected, first failed check: phase mismatch."	3 Actions Connect, inspect dependency, preview, cancel.
--	--	---

Implementation contract

The semantic record is authoritative for nonvisual navigation; alt text alone is insufficient.

Queryable graph summaries

Provide commands such as scene summary, current object, neighbors, dependencies, route history, failed check and available repair.

1

Concise mode

Only changed or actionable facts.

2

Detailed mode

Full relation and provenance trace.

3

Landmarks

Bench, rule shelf, candidate mat, exit.

Implementation contract

Queries use stable IDs so replay and focus remain coherent.

Captions and narration

All speech and meaningful audio have captions; essential text can be narrated.

1

Speaker and source

Identify who or what produced the sound.

2

Timing

Keep captions long enough and avoid covering controls.

3

Control

Pause, replay, speed and verbosity are adjustable.

Implementation contract

Narration and captions are synchronized to semantic events rather than only video time.

Sound cues have equivalents

A cue may be sound, visible pulse, caption, haptic pattern or tactile device output - never sound alone.

1 Name the event "Guard crossed."	2 Name urgency Informational, warning or stop.	3 Let the child choose Turn channels on or off independently.
---	--	---

Implementation contract

Procedural audio from KC 3.9 remains presentation, not authority.

Haptics are optional and named

Vibration patterns can reinforce state but must be predictable, adjustable and replaceable.

1

Requirement 1

No surprise vibration.

2

Requirement 2

Intensity and duration are limited.

3

Requirement 3

A visual/spoken/tactile alternative exists.

Implementation contract

Haptic feedback never carries unique safety or puzzle information.

Text and reading

Use plain language, meaningful headings, short paragraphs, selectable text, scalable type and reflow.

1

Child layer

Simple words and examples.

2

Builder layer

More precise concepts.

3

Formal layer

Original substrate terms and contracts.

Implementation contract

Reading level can change without changing object identity or rule consequence.

Reduced motion

Allow static camera, no shake, no parallax, slower transitions, reduced particles and instant state changes.

1

Requirement 1

Show before/after.

2

Requirement 2

Preserve timing in text.

3

Requirement 3

No flashing essential cue.

Implementation contract

Motion setting is available before the first animated scene.

Sensory forecast

Before a room, expose sound, light, motion, touch/haptic, reading, timing and social-demand descriptors.

1

Intensity

none, low, medium, high.

2

Control

adjustable, avoidable or required.

3

Alternative

available route.

Implementation contract

Forecast values are authored and tested, not guessed at runtime from the child.

Constructive error explanations

A rejected action names the first failed rule and an available repair.

1

Not “wrong.”

Say “not yet: outside support.”

2

Show evidence

Name residual or mismatch in suitable language.

3

Offer next step

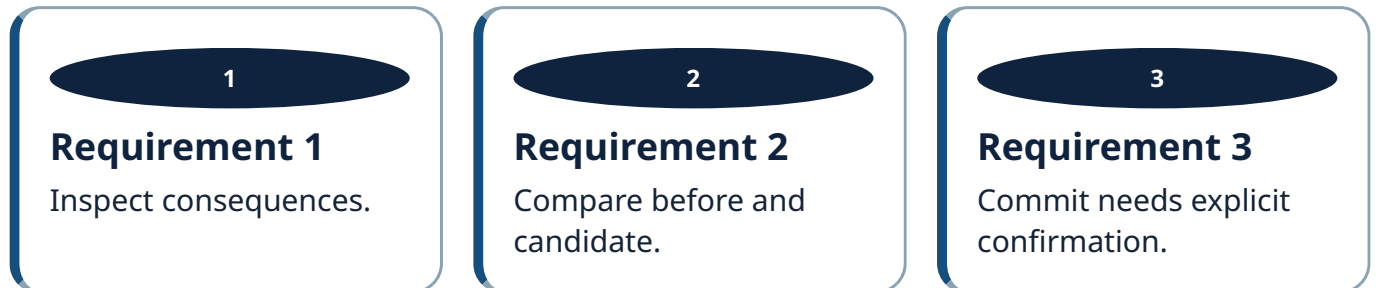
Move closer, change phase, inspect dependency or cancel.

Implementation contract

Error explanation is available through every output route.

Candidate preview

Preview clearly separates proposed and real state through label, border pattern, spoken cue and semantic status.



Implementation contract

Candidate state cannot silently trigger irreversible gameplay or analytics.

Undo and replay

Undo restores a previous valid state where the puzzle allows it; replay explains accepted changes even when undo is not allowed.

1

Requirement 1

Named checkpoint.

2

Requirement 2

Accessible event
summary.

3

Requirement 3

First divergence report
for technical debugging.

Implementation contract

Undo and replay controls are reachable through all access routes.

Save and return

Save focus, candidate, route and communication state without requiring the child to reconstruct context.

1

Return summary.

Continue, review or leave.

2

No streak penalty.

No repeated first-run calibration.

Implementation contract

KC 3.9 persistence can store canonical game state; access preferences remain separately scoped.

Multiplayer and co-play

Accessible routes retain equal authority in local and networked play.

1

Requirement 1

Turn ownership is explicit.

2

Requirement 2

Partner actions identify the child they relay for.

3

Requirement 3

Latency accommodation does not lower reward.

Implementation contract

Authoritative commit records whose confirmed intent produced each proposal, without exposing private access data.

Offline and low-end support

Core access works without a network, cloud speech service or high-end device.

1

Requirement 1

Self-contained build.

2

Requirement 2

Text and captions local.

3

Requirement 3

Reduced effects and low-compute view.

Implementation contract

KC 3.9 provides a self-contained HTML5 direction; cloud-only access is not acceptable for essential play.

Privacy and child data

Collect the minimum needed for function and evidence.

1

Requirement 1

No emotion recognition.

2

Requirement 2

No raw gaze retention by default.

3

Requirement 3

No advertising profile from access settings.

Implementation contract

Children and caregivers can inspect, export or delete stored access information.

Accessibility lint

Authoring tools automatically flag missing semantic names, single-channel clues, drag-only actions, inaccessible focus, missing captions and unequal rewards.

1

Requirement 1

Block severe defects.

2

Requirement 2

Link the exact object and action.

3

Requirement 3

Allow documented review, not silent ignore.

Implementation contract

Lint complements child testing; it does not certify accessibility alone.

Release gates

A build cannot ship when an essential route fails the eight-stage envelope or parity tests.



Implementation contract

Release evidence includes test cases, participant feedback, known limitations, defect owners and retest results.

Physical play principles

Physical play teaches the same concepts through shape, texture, position, connection, order and replay.

1

Same semantic IDs

A physical SUPPORT tile and digital SUPPORT object share a code.

2

Multiple perception routes

Word, letter, shape, border pattern, tactile mark and spoken name.

3

Child control

Pieces, partners and schedules preserve STOP, WAIT, NOT THAT and BREAK.

Prototype boundary

All designs are prototype scaffolds pending child co-design, material review, age grading and toy-safety assessment.

Accessible rule tile anatomy

A rule tile combines redundant identity cues and typed connection points.



letter + word + border pattern + tactile mark + typed socket

1

Face

Large word, letter code and optional symbol.

2

Edge

Distinct border pattern and tactile notch.

3

Socket

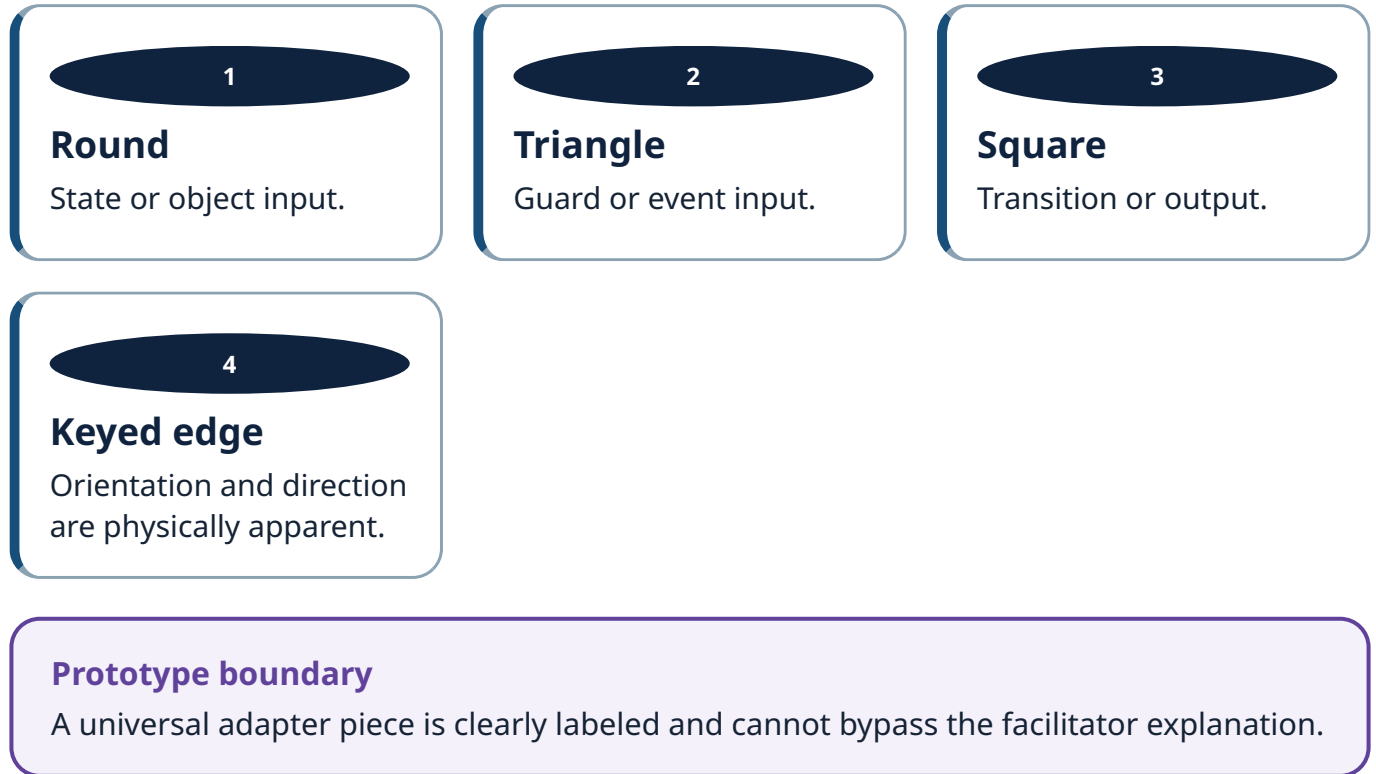
Shape-coded connector that rejects incompatible links mechanically where possible.

Prototype boundary

No essential distinction relies on color or tiny print.

Typed sockets

Socket shape teaches domain and codomain without requiring reading.



Support mat

A raised or high-contrast region shows where a rule is relevant.

 Inside/outside Use texture and edge height.	 Movable window Slide a support frame over the board.	 Nonvisual query Trace the boundary from a home marker.
--	---	---

Prototype boundary

The support mat prunes candidates; it does not decide compatibility.

Compatibility edges

Matching edge codes show whether supported pieces may connect.



Requirement 1
Shape code.



Requirement 2
Texture or notch code.



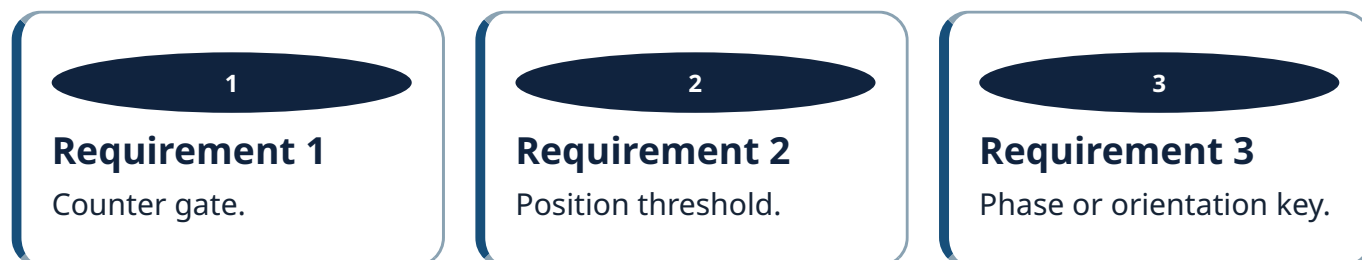
Requirement 3
Word/letter label.

Prototype boundary

A mismatch can be explored safely and explained as “not yet: these edges do not match.”

Guard gate

A gate opens only when a visible or tactile condition is reached.

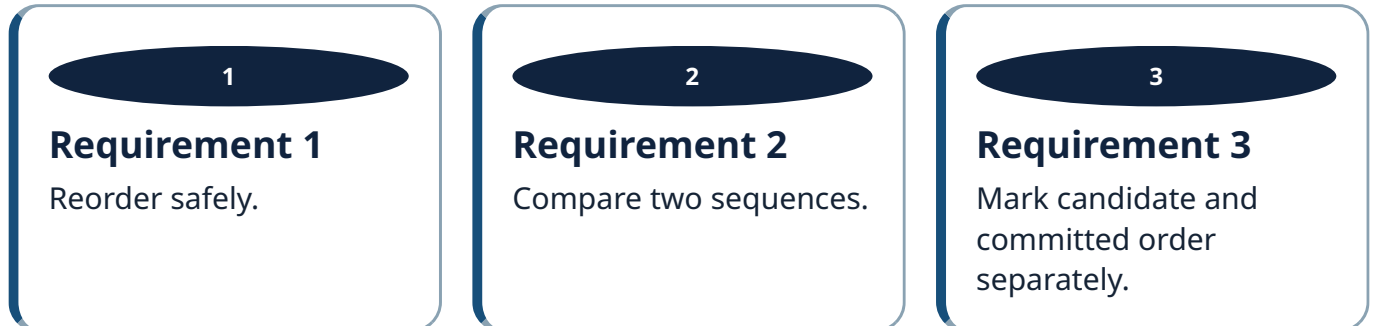


Prototype boundary

The gate has a clear candidate state and cannot snap unexpectedly onto a child's hand.

Operation-order rail

A rail makes sequence physical: cards are placed left to right and evaluated in that order.



Prototype boundary

MOVE then ROTATE and ROTATE then MOVE use the same cards but different rails.

Connector hinge

A hinge tile can join two pieces, pivot, or select between routes.

1

Requirement 1

Zero magnitude, real structure.

2

Requirement 2

Distinct pivot and connector forms.

3

Requirement 3

Guarded flip rather than loose accidental change.

Prototype boundary

The connector teaches that structural role is not the same as numerical size.

Topology track

A track can wrap, flip orientation or move between sheets while keeping the route continuous.

1

Requirement 1
Raised arrows.

2

Requirement 2
Orientation token.

3

Requirement 3
Entry and exit ports with matching IDs.

Prototype boundary
The model is a routing abstraction, not a claim that a physical toy is a literal Klein bottle.

Sheet overlays

Transparent, opaque or tactile overlays allow two states to share a coordinate while remaining distinct.

1

Requirement 1

Sheet A and B labels.

2

Requirement 2

Compatibility windows.

3

Requirement 3

Nontransparent option
for children who need
lower visual complexity.

Prototype boundary

A child can compare “same place” with “same identity” and “allowed to interact.”

Winding counter

A bead, dial or click wheel remembers how many loops occurred after phase repeats.

1

Requirement 1
One tactile click per loop.

2

Requirement 2
Parity mark for odd/even.

3

Requirement 3
Reset only by explicit rule.

Prototype boundary
Winding is lineage, not hidden score.

Constraint bars

Bars, clips or rails limit legal movement directions.

1

Requirement 1

Add a constraint.

2

Requirement 2

Remove one declared shackle.

3

Requirement 3

Compare movement before and after.

Prototype boundary

Use low-force parts and avoid pinch points. Constraint release does not demonstrate chaos by itself.

Replay beads and lineage chain

Each accepted change adds a bead or link with a rule code.

1

Requirement 1

Stable start marker.

2

Requirement 2

One bead per committed event.

3

Requirement 3

Branch and merge connectors.

Prototype boundary

A child can replay by moving along the chain; rejected candidates use a separate temporary tray.

Object cues and tactile schedule

Objects announce the activity and sequence before play.

1

Requirement 1
Rule tile = rule game.

2

Requirement 2
Soft ring = break.

3

Requirement 3
Replay bead = review.

Prototype boundary
Meanings are agreed with the child and remain consistent across settings.

Low-force and mounting options

Pieces work on a table, lap tray, vertical board, non-slip mat or mounted switch surface.

1

Requirement 1

Large grasp area.

2

Requirement 2

Low insertion force.

3

Requirement 3

Magnetic and magnet-free prototypes.

Prototype boundary

No child must stabilize the whole board while operating a piece.

Tactile and large-print variants

Offer large pieces, braille-ready labels, tactile line diagrams, strong contrast and simplified layouts.

1

Requirement 1

Tactile symbols require user testing.

2

Requirement 2

Braille placement does not cover other cues.

3

Requirement 3

Provide orientation marks.

Prototype boundary

Braille and tactile graphics require qualified review; placeholders are not validated codes.

Quiet and magnet-free variants

A quiet kit uses soft stops and nonclicking markers. A magnet-free kit uses keyed mechanical connections.

1

Requirement 1

No surprise snap.

2

Requirement 2

No loose electronic sound.

3

Requirement 3

Material choices documented.

Prototype boundary

Variants preserve the same connection logic and reward.

Household prototype route

Early learning prototypes can use card, foam, hook-and-loop, string, bottle caps and textured tape.

1

Requirement 1

Use oversized parts.

2

Requirement 2

Round sharp corners.

3

Requirement 3

Supervise and inspect wear.

Prototype boundary

Household prototypes are adult-prepared learning aids, not certified toys.

Product development placeholders

Every toy concept includes spaces for age range, part size, material, force, cleaning, tactile code, mounting, packaging and test evidence.



Physical safety and regulatory boundary

The substrate can define a learning mechanic, but it cannot certify a product.

1

Requirement 1

Choking and ingestion.

2

Requirement 2

Pinch, entrapment and sharp edges.

3

Requirement 3

Magnets, batteries and cords.

4

Requirement 4

Flammability and chemical safety.

5

Requirement 5

Cleaning and infection control.

6

Requirement 6

Age grading and supervision.

Prototype boundary

Qualified toy-safety, accessibility, clinical and regulatory review is required before sale or deployment.

A ten-minute lesson

Goal: experience support and compatibility without reading or speed.

1

1 minute

Show STOP and the child's answer route.

2

2 minutes

Place two rule tiles and a support mat.

3

3 minutes

Find which tile is inside support.

4

2 minutes

Test whether edges match.

5

2 minutes

Preview, confirm and replay one change.

Facilitator rule

Finish while the child still has agency and energy; a complete curriculum is not required in one session.

A thirty-minute lesson

Goal: build a complete six-question trace with breaks.

1

Welcome

Access check and sensory forecast.

2

Explore

Definitions, dependencies and support.

3

Build

Compatibility, guard and candidate.

4

Commit

Confirm and change.

5

Replay

Lineage chain and reflection.

Facilitator rule

Offer micro-breaks and an early FINISHED option at every transition.

Cooperative classroom roles

Roles let children contribute through different strengths without ranking them.

1

Observer

Notices objects and support.

2

Matcher

Checks compatibility.

3

Guard Keeper

Watches the boundary.

4

Committer

Confirms the group choice.

5

Replay Guide

Retells or reconstructs changes.

Facilitator rule

Roles rotate only with consent; every role can be performed by speech, AAC, touch, switch, gaze or partner-assisted choice.

Family play

Use ordinary routines to reveal substrate ideas.

1

WHERE

Which shelf can hold this object?

2

MATCH

Which lid fits?

3

WHEN

When is the cup full?

4

CHANGE

What happens after the timer?

5

REMEMBER

Put one bead for each accepted change.

Facilitator rule

Keep the activity playful and stop before it becomes a lesson demand.

Accessible assessment

Assess the concept through multiple expressions, not reading speed, speech or fine-motor performance.

1

Show

Arrange tiles or point.

2

Tell

Speak, sign or use AAC.

3

Do

Choose a candidate and repair it.

4

Replay

Explain what changed in any channel.

Facilitator rule

Record evidence of understanding separately from access effort and adult support.

Child co-design

Children shape rules, access routes, language, symbols, sensory levels and repair behavior.

1

Accessible invitation.

Real choices and the right to decline.

2

Prototype in the child's route.

Pay or recognize contribution fairly.

3

Requirement 3

Show what changed because of feedback.

Facilitator rule

Co-design is not a one-time usability test after decisions are fixed.

Facilitator behavior

The adult makes access possible without becoming the player.

1

Requirement 1

Offer, do not force.

2

Requirement 2

Wait, do not fill silence.

3

Requirement 3

Confirm, do not assume.

4

Requirement 4

Describe, do not quiz.

5

Requirement 5

Repair, do not blame.

6

Requirement 6

Stop when the child stops.

Facilitator rule

Training includes communication partner fidelity and touch-consent practice.

Safeguarding and assent

Participation is voluntary and continuously revocable.

1

Explain activity and data.

Provide STOP and exit before consent.

2

Watch for child-specific distress or withdrawal signals.

Never withhold ordinary support to obtain participation.

Facilitator rule

Research or evaluation requires appropriate ethical, legal and organizational review.

Combined-access test matrix

Test realistic combinations instead of isolated checkboxes.

1

Requirement 1

Blind + switch.

2

Requirement 2

Low vision + gaze.

3

Requirement 3

AAC + partner scan.

4

Requirement 4

Deafblind + fatigue.

5

Requirement 5

Reduced motion + screen reader.

6

Requirement 6

One hand + captions + cognitive simplification.

Facilitator rule

Record route discovery time, successful stop, confirmation, repair, return and concept parity.

Child-defined metrics

Ask whether the experience worked for the child.

1

Requirement 1

Could I find a route?

2

Requirement 2

Could I stop?

3

Requirement 3

Did the game understand me?

4

Requirement 4

Could I fix a misunderstanding?

5

Requirement 5

Could I learn the same idea?

6

Requirement 6

Could I return comfortably?

Facilitator rule

Metrics are not used to rank children; they evaluate the design.

Release evidence pack

Every release carries traceable accessibility evidence.

1

Requirement 1

Requirement map.

2

Requirement 2

Automated lint report.

3

Requirement 3

Assistive-technology test cases.

4

Requirement 4

Child and family feedback.

5

Requirement 5

Known limitations and mitigations.

6

Requirement 6

Defect owner and retest date.

Facilitator rule

Standards crosswalks support evidence but do not replace participatory testing.

Roadmap and kill criteria

Stop or simplify a feature when it cannot preserve child agency, concept parity or safe use.

1

Kill drag-only, speech-only and color-only gates.

Kill unconfirmed partner control.

2

Kill inaccessible STOP, help, settings or return.

Kill unequal reward or segregated endings.

3

Requirement 3

Kill privacy-invasive gaze or emotion tracking by default.

Facilitator rule

A smaller accessible mechanic is better than a larger inaccessible claim.

Child usability outcome

A child-usable system lets a child achieve a meaningful goal effectively, with manageable effort and a felt sense of control, through a route that works in that session.



1

Effective

The child can make the intended choice, understand the result and reach the same concept outcome as other routes.

2

Efficient enough

The route avoids needless steps, repeated setup, excessive scanning, precision demand and adult mediation.

3

Understandable

Purpose, current state, available choices and likely effects stay visible or queryable.

4

Repairable

Misunderstandings, wrong actions and system faults can be explained, undone and resumed without blame.

5

Satisfying and dignified

The child can experience play, agency, competence and belonging without access tools lowering status.

6

Learnable and memorable

Stable patterns, words and state transitions reduce relearning and support return after interruption.

Usability gate

An accessible control that a child cannot discover, understand, confirm, repair or return to is not yet usable.

Developmental routes without age gates

Use observed readiness, communication, experience, interests, energy and access needs. Age may inform examples, but never becomes a gate to the full concept.

1

Explore route

Two choices, concrete objects, direct cause and effect, one question at a time and immediate replay.

2

Build route

Connect rules, compare order, repair mismatches and predict one next state.

3

Explain route

Describe a reason through speech, AAC, pointing, sorting, demonstration, replay or partner relay.

4

Design route

Create a rule, declare dependencies, set a guard, test edge cases and review lineage.

5

Move between routes

A child may use Explore for a new concept and Design for a familiar one in the same session.

6

No lesser track

Every route teaches the same substrate truth; only representation, scaffolding and depth change.

Usability gate

Do not use reading level, speech, hand control, eye contact or chronological age as a proxy for conceptual capacity.

Child journey and friction map

Design the whole journey, not only the central puzzle. Most exclusion appears before the first choice, during interpretation, at error repair or when returning after a break.



1

Prepare friction

Unknown sound, motion, duration, people, touch or login demands.

2

Perceive friction

Color-only state, audio-only clues, crowded layouts or inaccessible relations.

3

Understand friction

Unclear purpose, unfamiliar icons, hidden rules and too many choices.

4

Act friction

Drag-only, hold-only, timed, precision or simultaneous-control requirements.

5

Confirm and repair friction

Silent commits, vague errors, lost focus, irreversible mistakes or adult override.

6

Pause and resume friction

Progress loss, missing return summary, forgotten scan position or memory tests.

Usability gate

Every release test must walk at least one representative child route through all eight stages.

The first minute and first success

The first encounter establishes whether the child owns the world. Setup must serve play rather than become a hidden qualification test.

1

No wrong door

Any first input route leads to a safe route chooser or first choice, not a dead end.

2

Control bar first

STOP, WAIT, HELP, UNDO, QUIET and BREAK are visible or announced before the first effect.

3

Two meaningful choices

Avoid a tutorial wall. Present two choices with clear effects and an ANOTHER route.

4

Safe candidate

The first action uses TRY and can be canceled without consequence.

5

Immediate confirmation

The system repeats the interpreted choice before commitment.

6

First success

The child completes one real world change before account creation, optional calibration or advanced settings.

Usability gate

Block release when a child must pass a reading, speech, gesture, calibration or tutorial quiz before making a meaningful choice.

Choice architecture and progressive disclosure

Usability improves when each choice is meaningful, distinguishable and small enough to compare. More is not automatically more autonomous.

1

Begin with two

Use two primary choices plus NOT SURE, ANOTHER or STOP as needed.

2

Reveal on request

More rules, detail and advanced controls appear when the child asks or the current task requires them.

3

Keep effects visible

Each choice says what object it affects and the likely result.

4

Avoid privileged YES

YES, NO, NOT SURE and CHANGED MIND receive equal visual and scan priority.

5

Preserve rejected choices

A NO or NOT THAT remains a valid decision and does not trigger persuasion.

6

Remember preference, not assumption

Offer prior choices as shortcuts while keeping change easy and visible.

Usability gate

Do not increase choice count to simulate freedom when it increases search, memory or scan burden without adding meaningful agency.

Information architecture and predictable navigation

A child should always know where they are, what is active, what can happen next and how to leave.

1

Stable places

HOME, PLAY, RULES, TRY, REAL, REPLAY and SETTINGS keep one meaning across rooms.

2

Current location

Show or announce room, step, candidate and last accepted change.

3

Multiple ways to find

Provide direct links, sequential navigation, contents, search or query summaries where relevant.

4

No focus traps

Keyboard, switch, screen-reader and gaze focus can leave every panel and dialog.

5

Return focus predictably

After closing help, undoing or returning from break, focus goes to the relevant object or prior scan position.

6

Consistent help

HELP appears in the same logical place and opens the same kind of ladder.

Usability gate

Navigation labels describe purpose, not visual position alone. "Next rule" is usable; "button on the right" is not sufficient.

Vocabulary and icon system

Use one stable state vocabulary and redundant cues. Symbols support meaning but do not replace tested language or individualized communication.

TRY candidate circle	CHECKING verifying diamond	READY can commit triangle	NOT YET repair clue square	REAL committed hexagon	REMEMBERED lineage ring
-----------------------------------	---	--	---	-------------------------------------	--------------------------------------



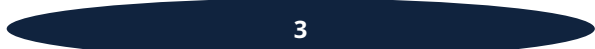
Core child words

WHERE, MATCH, WHEN, CHECKED, CHANGE and REMEMBER map to the real substrate sequence.



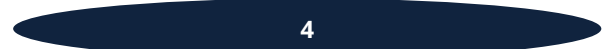
World states

TRY, CHECKING, READY, NOT YET, REAL and REMEMBERED remain consistent across toys and screens.



Redundant coding

Combine word, short letter, shape, border pattern, position, optional sound and optional tactile mark.



No color-only meaning

Color decorates or reinforces; it never carries the only distinction.



Familiar controls

Use conventional pause, undo, close and help patterns where they are genuinely familiar to the child.



Test symbols locally

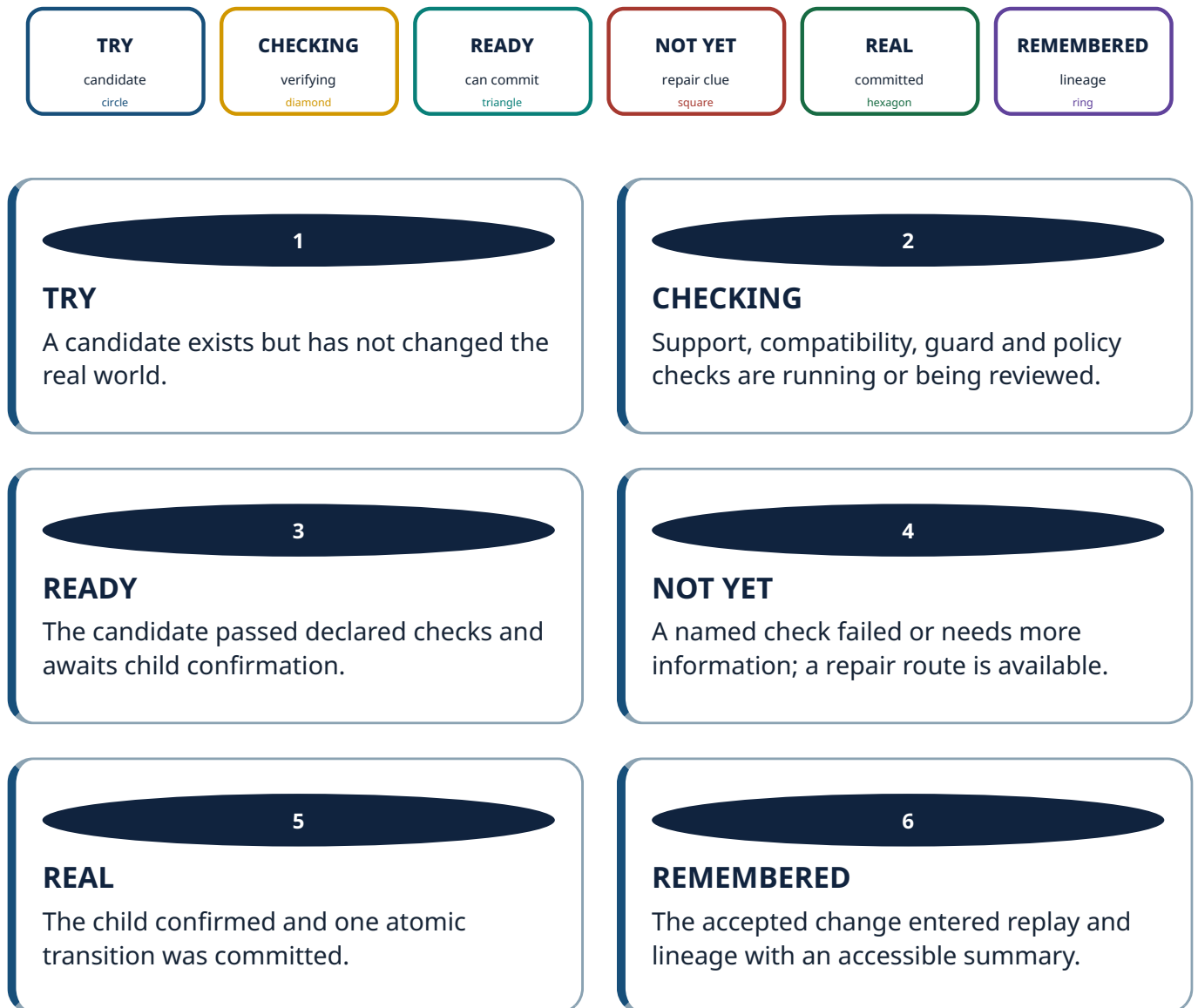
AAC, tactile, sign and cultural symbol choices require user and specialist validation.

Usability gate

Never invent a supposedly universal tactile, AAC or sign-language symbol set without participatory validation.

Feedback state machine

Every action receives immediate, consistent feedback about what the system heard, what it is checking and whether the world changed.



Usability gate

Avoid generic “wrong,” “invalid” or success-only color flashes. State feedback must be semantic, persistent enough to inspect and available through all active routes.

Hint ladder and help without takeover

Help should reduce the current barrier while preserving the child's final decision and ability to step back.



1

Level 1 - Goal

Remind what the child is trying to make happen.

2

Level 2 - Relation

Point to the relevant WHERE, MATCH, WHEN or order relation.

3

Level 3 - One step

Show one possible next action without committing it.

4

Level 4 - Together

Perform one reversible step with the child through the agreed assistance route.

5

Level 5 - Solution

Show a complete solution only after explicit choice; preserve replay and the option to try independently later.

6

Repair the help

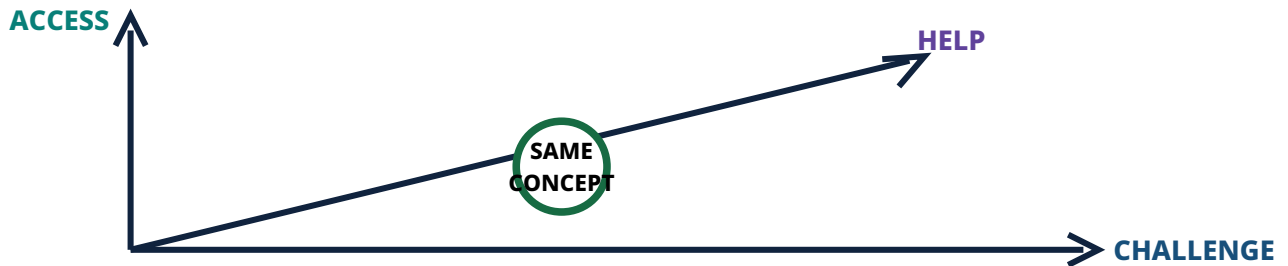
The child can say TOO MUCH, DIFFERENT WAY, AGAIN or CLOSE HELP.

Usability gate

Hints never auto-commit, remove reward, change difficulty, disclose private access data or become compulsory after waiting.

Difficulty, access and help are separate axes

A child may need a complex concept with a simple presentation, or a simple concept with rich exploration. Do not collapse these dimensions.



Changing an access route must not secretly lower the concept or reward.

1

Challenge axis

Concept depth, number of interacting rules, prediction horizon and proof demand.

2

Access axis

Perception, input, communication, timing, layout, sensory and assistive-technology routes.

3

Help axis

Explanation, hints, examples, co-play and partner support.

4

Same outcome

Equivalent concept outcomes receive equivalent credit and endings.

5

No access penalty

Captions, switches, AAC, narration, reduced motion and hints are not cheats.

6

Child choice

The child may change any axis independently during a session.

Usability gate

Any implementation that lowers score, concept status, role or ending because of an access route fails parity.

Session pacing, fatigue and return

Pacing is part of usability. The system must support attention shifts, pain, fatigue, sensory change and interrupted schedules without turning them into failure.

1

Session menu

Offer one-action, 3-minute, 8-minute, 15-minute and open play options.

2

No hidden countdown

Time estimates inform; they do not pressure unless timing is the chosen learning goal.

3

Micro-checkpoints

Save after every accepted transition and before major sensory or navigation changes.

4

Effort forecast

Name likely reading, scanning, motor, social and sensory effort.

5

Safe pause

Pause scanning, sound, animation and input interpretation without committing a pending action.

6

Return summary

Restore access route, focus, candidate and a concise “you were here” summary.

Usability gate

A child must not lose progress, reward or role because an access route is slower or a break is needed.

Motivation, reward and fairness

Motivation grows from meaningful choice, understandable challenge, repair, curiosity, cooperation and ownership - not from pressure or inaccessible competition.

1

Reward understanding

Recognize correct relation, clear repair, useful explanation, creative rule and cooperative contribution.

2

No speed default

Do not rank by response speed unless timing is an explicitly chosen and accessible concept.

3

No speech premium

Spoken explanations are one route among many.

4

Celebration menu

Offer visual, auditory, tactile, social, quiet or no-celebration choices.

5

Avoid streak pressure

Breaks, route changes and NOT SURE do not destroy progress streaks.

6

Own the goal

Let the child select puzzles, characters, themes, session length and next activity where possible.

Usability gate

Do not use dark patterns, forced continuation, loss-framed countdowns or adult-facing rewards that pressure the child to keep playing.

Digital interaction requirements

The digital game should expose one semantic world through multiple input and output adapters, with generous targets and alternatives to precision gestures.

1

Targets

Meet the WCAG 2.2 minimum target-size rules and use a larger child preset by default; spacing and testing matter as much as nominal size.

2

Alternatives to dragging

Every drag has tap-select-place, directional, keyboard, switch, gaze or partner-assisted equivalent.

3

Visible and semantic focus

Focus is visible, not obscured, announced, stable and restored after dialogs, help and errors.

4

No hold or simultaneous requirement

Provide toggle, sequential and one-handed routes.

5

Adjustable timing

Scanning, dwell, repetition and timeout behavior are adjustable and pausable.

6

Performance and offline use

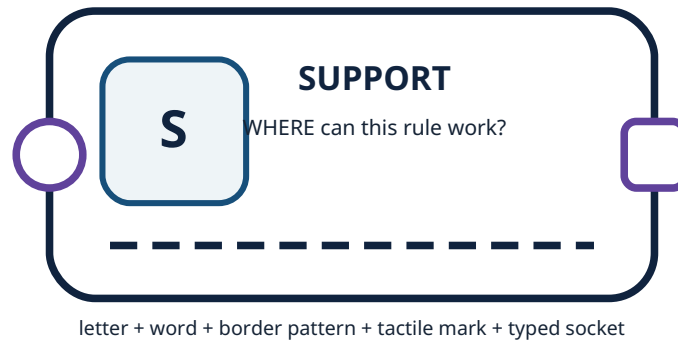
Core play, saves, access settings and explanations work without a network and on lower-performance targets.

Usability gate

Block release when any essential action has only a drag, hold, double-tap, rapid repeat, precision aim or simultaneous multi-button route.

Physical interaction requirements

Physical prototypes must be graspable, stable, low-force, distinguishable and adaptable without making the child hold pieces in one prescribed way.



1

Large, stable pieces

Use proportions suited to the target age and access route; provide anti-slip bases, trays, mounts or vertical boards.

2

Low force

Connectors, gates and rails operate with minimal force and can be used one-handed or with a tool.

3

Redundant identity

Word, shape, border, texture and socket type remain consistent with the digital object.

4

Quiet and nonmagnetic variants

Offer alternatives where magnets, clicks, loose parts, vibration or noise are unsuitable.

5

Hand-under-hand option

Support consensual exploration without forced hand-over-hand movement.

6

Safety placeholders

Material, choking, pinch, magnet, allergy, cleaning, durability and age-grading review remain mandatory before production.

Usability gate

These are educational prototype requirements, not manufacturing drawings or toy-safety certification.

Twelve concept microgames

Each substrate idea has a short physical game and a semantically equivalent digital room. Any microgame can be one action or part of a longer campaign.

 WHERE Hunt Sort what can reach the marked area.	 MATCH Workshop Connect compatible typed sockets.	 Guard Gate Distinguish near, touch and cross.
 Checker Trail Run a candidate through three checks.	 TRY-to-REAL Preview, confirm and commit one change.	 Replay Beads Reconstruct accepted changes in order.
 Rule Recipe Build a definition card and its needs.	 Order Rail Compare two non-commuting orders.	 Flip Track Follow orientation through a wrap.
 Loop Counter Separate phase from winding.	 Seed Garden Rebuild a bounded branch world.	 Honest Code Name what a compressed record keeps and loses.

Usability gate

A microgame succeeds when the child demonstrates the relation through any valid route, not when a preferred behavior or body movement is reproduced.

Physical-to-digital parity

Children should not have to relearn the conceptual grammar when moving from toys to tabletop play or a screen.

1

Stable names

The same rule, state and action words appear across media.

2

Stable identities

A rule tile ID, socket type and relation map to the same semantic object digitally.

3

Stable sequence

WHERE, MATCH, WHEN, CHECKED, CHANGE and REMEMBER stay in the same order.

4

Stable state boundary

TRY and REAL remain distinct in every medium.

5

Stable replay

Physical beads and digital event records represent the same accepted sequence.

6

Transfer test

A child can solve a familiar concept in a new medium without a new hidden rule.

Usability gate

Visual resemblance alone is not parity. Test semantic transfer, child explanation and error repair across media.

Child usability research protocol

Research must include children who use the intended routes and must be accessible, ethical, safe, meaningful and capable of changing the product.

1

Assent and stop

Explain the activity accessibly. Honor assent, dissent, STOP, withdrawal and changing minds.

2

Accessible participation

Offer AAC, sign, tactile, object, switch, gaze, partner and low-reading methods.

3

Representative combinations

Test combined and fluctuating routes, not only isolated impairment categories.

4

Observe the system

Record where interface, wording, timing, layout, partner behavior or physical design created friction.

5

Do not diagnose

Usability testing studies the product and route; it is not a clinical, intelligence or compliance assessment.

6

Close the loop

Tell children what changed because of their participation in an accessible form.

Usability gate

A design cannot claim child-centered usability when children with relevant access needs were excluded from research or their feedback could not alter release decisions.

Measurement, release gates and evidence pack

Use a balanced evidence set: observed task outcomes, child-reported experience, assistive-technology behavior, partner fidelity and concept transfer.

1

Start metrics

Time and steps to first meaningful choice; route discovery; STOP discovery; setup abandonment reason.

2

Choice metrics

Correct interpretation, confirmation success, accidental activation, scan burden and adult takeover rate.

3

Repair metrics

NOT THAT and UNDO success, explanation quality, state restoration and emotional safety.

4

Return metrics

Access-route, focus, candidate, checkpoint and summary restoration after interruption.

5

Learning metrics

Concept demonstration, transfer across media, prediction, explanation and later reconstruction.

6

Experience metrics

Child-reported comfort, effort, joy, ownership, preferred celebration and willingness to return.

Usability gate

Release is blocked by inaccessible essential actions, hidden commits, lost state, forced touch, inferred consent, adult substitution, lower accessible rewards, inaccessible help or unrepairable interpretation errors.

Printable core communication board

Point, look, touch, sign, scan or have a partner name each cell. Keep this board visible throughout play.

STOP**GO****WAIT****HELP****AGAIN****FINISHED****YES****NO****NOT SURE****NOT THAT****CHANGED MIND****BREAK****QUIET****DIM****STILL****BIGGER****READ****TOUCH****SWITCH ROUTE**

Prototype sheet - print at 100% or enlarge. Add tactile or symbol supports only through user-tested conventions.

Printable YES / NO / NOT SURE / CHANGED MIND

Four equal-size choices. Do not place YES in a visually privileged position.

YES

NO

NOT SURE

CHANGED MIND

Printable STOP / HELP / WAIT / BREAK

Safety and control words remain available before, during and after every action.

STOP

HELP

WAIT

BREAK

Prototype sheet - print at 100% or enlarge. Add tactile or symbol supports only through user-tested conventions.

Printable FIRST / THEN / NEXT / FINISHED

Place one object, word or symbol under each heading. The child may reorder or remove steps.

FIRST

THEN

NEXT

FINISHED

Printable sensory forecast

Mark NONE, LOW, MEDIUM or HIGH and whether it can be changed, skipped or replaced.

SOUND

LIGHT

MOTION

TOUCH

READING

TIME

PEOPLE

BREAKS

Printable access-route chooser

Choose one or more routes for this session. A route can be changed without losing the puzzle.

LOOK**LISTEN****TOUCH****READ****SPEAK****SIGN****AAC****SWITCH****GAZE****HEAD****KEYS****PARTNER**

Printable My Access Passport

Share only what the child chooses. Do not add diagnosis labels unless the child or family explicitly wants them included.

MY YES

MY NO

MY STOP

MY NOT SURE

WHAT HELPS

WHAT TO AVOID

HOW TO WAIT

HOW TO RETURN

Prototype sheet - print at 100% or enlarge. Add tactile or symbol supports only through user-tested conventions.

Printable partner scan record

Use to train and review partner fidelity. The child is not being scored.

CHOICES	SCAN ORDER
CHILD SIGNAL	CONFIRMED CHOICE
REPAIR USED	WAIT TIME

Printable six-question cards

Cut into large cards. Add tactile marks or object cues only after user testing.

**WHERE?
SUPPORT**

**MATCH?
COMPATIBILITY**

**WHEN?
GUARD**

**CHECKED?
VERIFIED**

**CHANGE
TRANSITION**

**REMEMBER
LINEAGE**

Prototype sheet - print at 100% or enlarge. Add tactile or symbol supports only through user-tested conventions.

Printable blank rule tile

Use large print and one idea per tile. Add a distinct border pattern and typed socket sketch.

RULE NAME

LETTER CODE

WHAT IT NEEDS

WHEN IT RUNS

WHAT IT CHANGES

HOW IT IS REMEMBERED

Printable dependency cable labels

Attach labels to string, pipe cleaners or card connectors. Keep loose-part safety in mind.

NEEDS

PROVIDES

BEFORE

AFTER

SAME TYPE

DOES NOT MATCH

OPTIONAL

REQUIRED

Prototype sheet - print at 100% or enlarge. Add tactile or symbol supports only through user-tested conventions.

Printable operation-order rail

Place rule cards in order. Compare two rails without changing the card identities.

1 FIRST

2 SECOND

3 THIRD

4 FOURTH

5 PREVIEW

6 COMMIT

Prototype sheet - print at 100% or enlarge. Add tactile or symbol supports only through user-tested conventions.

Printable candidate and real-world mat

Candidate pieces never move to REAL WORLD without explicit confirmation.

TRY / CANDIDATE

CHECKS

CONFIRM

REAL WORLD

CANCEL

REPLAY

Prototype sheet - print at 100% or enlarge. Add tactile or symbol supports only through user-tested conventions.

Printable repair ladder

Use after a misunderstanding. Never require the child to apologize for access failure.

STOP

**WHAT DID YOU THINK I
CHOSE?**

YES / NO / NOT THAT

SHOW AGAIN

CHOOSE ANOTHER ROUTE

CONFIRM

Printable topology track

Draw or build a track. Use arrows and tactile orientation marks. Record the wrap as an event.

ENTRY

WRAP

FLIP

SHEET A

SHEET B

EXIT

Prototype sheet - print at 100% or enlarge. Add tactile or symbol supports only through user-tested conventions.

Printable winding and replay beads

Thread or arrange large safe pieces. Keep winding count separate from phase position.

START

LOOP 1

LOOP 2

LOOP 3

BRANCH

MERGE

CHECKPOINT

FINISH

Prototype sheet - print at 100% or enlarge. Add tactile or symbol supports only through user-tested conventions.

Printable cooperative role cards

Any role can use any communication or input route. A child may pass or share a role.

OBSERVER

MATCHER

GUARD KEEPER

COMMITTER

REPLAY GUIDE

ACCESS CHECKER

Printable return-to-play card

Use after a break, device loss or interrupted session. Do not test memory.

YOU WERE HERE

LAST ACCEPTED CHANGE

CURRENT CANDIDATE

YOUR ACCESS ROUTE

CONTINUE

REVIEW

LEAVE

Printable My Play Menu

Choose a session size before play. The child can change it or finish at any time without losing progress.

ONE ACTION

3 MINUTES

8 MINUTES

15 MINUTES

OPEN PLAY

STOP NOW

Printable two-choice board

Begin with two meaningful choices and equal access to ANOTHER, NOT SURE, NOT THAT and STOP.

CHOICE A

CHOICE B

ANOTHER

NOT SURE

NOT THAT

STOP

Printable feedback state cards

Use the same state words across tabletop, toy and digital versions. Add redundant shape, border and tactile cues after testing.

TRY

CHECKING

READY

NOT YET

REAL

REMEMBERED

Printable hint ladder

The child chooses the level. A hint never commits an action or removes reward.

1 REMIND GOAL

2 POINT TO RELATION

3 SHOW ONE STEP

4 DO ONE TOGETHER

5 SHOW SOLUTION

LESS HELP

DIFFERENT WAY

CLOSE HELP

Printable challenge dial

Change challenge, access and help independently. The concept and reward remain equivalent.

EXPLORE

BUILD

EXPLAIN

DESIGN

KEEP MY ACCESS

CHANGE HELP

Printable play-again or finish board

Offer a predictable ending. FINISH is a valid choice and must not trigger persuasion or loss.

AGAIN

REVIEW

NEW GAME

SAVE

BREAK

FINISH

Printable microgame deck

Cut into concept cards. Select one card for a micro-session or combine them into a campaign.

WHERE HUNT	MATCH WORKSHOP	GUARD GATE
CHECKER TRAIL	TRY TO REAL	REPLAY BEADS
RULE RECIPE	ORDER RAIL	FLIP TRACK
LOOP COUNTER	SEED GARDEN	HONEST CODE

Printable child usability observation sheet

Record product and route behavior, not a score for the child. Include the child's own report and what will change in the design.

START	CHOICE	CONFIRM
REPAIR	STOP	RETURN
EFFORT	JOY	ADULT TAKEOVER
CONCEPT TRANSFER		

Source map and guidance crosswalk



DB-3.0 retains the suitable UGTS versions and adds a usability layer without changing their authority or evidence boundaries.

UGTS-GN 1.1

Canonical support, compatibility, guard, event, transition and lineage sequence; precision, memory and claim boundaries.

UGTS-KC 2.0

Patterns, fields, topology descriptors, kinematics, bounded dynamics and hybrid-event classification.

KC Two Hands 3.0

Stable scene identity, typed interaction, proposals, deterministic commit, checkpoints and replay.

UGTS-KC 3.6

First-class definitions, content addresses, dependency graphs, hinges and explicit operation order.

SCLP 3.6.2

Guard intervals, winding, reflective topology, constraint release, bounded grammar, finite keys and honest compression boundaries.

KC Elizabeth 3.9

Vector assets, named multi-device actions, deterministic fixed-step game world, persistence and offline HTML5 delivery.

SARA 3.6.3

Optional older-learner permission, public/private and authorization lesson. Secret search and transaction capabilities remain excluded.

Definition Breaker DB-2.0

Retained communication, functional access, substrate learning, digital, physical, teaching and printable reference baseline.

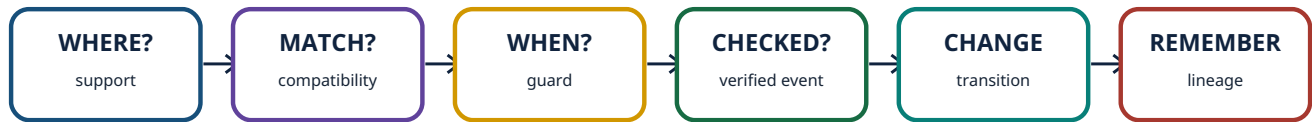
Current child-usability and accessibility guidance

Design and testing are informed by W3C WCAG 2.2, W3C Making Content Usable for People with Cognitive and Learning Disabilities, CAST UDL Guidelines 3.0, UNICEF child participation and disability-inclusive digital design guidance, and National Center on Deafblindness principles such as “do with, not for,” hand-under-hand support, pacing and wait time.

w3.org/TR/WCAG22/ | w3.org/TR/coga-usable/ | udlguidelines.cast.org | unicef.org | nationaldb.org

These are guidance and testing references. This PDF does not claim product, educational, clinical, PDF/UA, toy-safety or regulatory certification.

Formal decision and validation boundary



Definition Breaker DB-3.0 is ready as a child-usable playbook and implementation standard built on one shared semantic world.

What is established in this edition

- A 40-page direct-use child playbook.
- Stable control, feedback, hint and return vocabularies.
- A physical-to-digital microgame bridge for the whole suitable substrate.
- Digital and physical usability gates, evidence metrics and printable prototypes.

What still needs real-world validation

- Participatory testing with diverse children and families.
- Independent assistive-technology, tagged-PDF, educational and safeguarding review.
- Braille, tactile, AAC, sign-language and language-profile validation.
- Toy safety, materials, age grading, manufacturing and regulatory assessment.

FORMAL DECISION

A child-usable Definition Breaker action must be discoverable, perceivable, understandable, operable, confirmable, repairable, pausable and resumable through a route that works for the child in that session. Access, help and challenge remain separate. The child retains decision authority. Every accepted action still follows support, compatibility, guard, verified event, transition and lineage.

SEE. CHOOSE. CHECK. PLAY. PAUSE. RETURN.
WHERE? MATCH? WHEN? CHECKED? CHANGE. REMEMBER.